Supplementary Information

Detailed computation of the structural, thermodynamic, radial-distribution, and interfacial observables for frozen-bias spherical water cavities

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Abstract

This document gives the complete analysis protocol used to compare water in spherical non-periodic cavities of radius 15, 20, and 30 Å (R15, R20, and R30) with a matched periodic reference. It defines the trajectory windows, molecular reconstruction, radial shell estimators, orientational and network observables, temporal fluctuation convention, frozen Gaussian mean-field reconstruction, boundary-safe and radially resolved radial distribution functions (RDFs), molecule- and volume-weighted histograms, practical-equivalence screen, experimental RDF comparison, and all numerical quality-control checks. The aim is to make every quantity in Supplementary Figures 1 to 28 reproducible from the archived trajectories and tables without relying on plotting-code defaults.

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1 Systems, trajectories, and analysis windows

1.1 Frozen-bias NPBC cavities and periodic reference

The three non-periodic systems contain only water and are confined within spherical cavities. The molecular mean-field biases were frozen before the validation trajectories used here. The final frozen profiles correspond to global optimizer updates 159 (R15), 212 (R20), and 188 (R30). The nominal target number density was

$$\rho_0 = 0.037242 \text{ molecules Å}^{-3}. \tag{1}$$

The actual geometric densities calculated from $N/(4\pi R^3/3)$ are given in table 1. The periodic reference, denoted PBC40, contains 2383 water molecules in a cubic cell of side 40.000 Å, and therefore has number density 0.037234375 Å^{-3}. It used the same MACE–OFF23 `small_2` potential as the reference production run.

Table 1: Systems used in the validation analysis. The outer-volume fractions are exact geometric fractions of a sphere.

System	R (Å)	Waters	ρ (Å^{-3})	outer 2 Å	outer 5 Å
R15	15	526	0.037207	0.349037	0.703704
R20	20	1248	0.037242	0.271000	0.578125
R30	30	4211	0.037233	0.186963	0.421296
PBC40	—	2383	0.037234375	—	—

For a spherical cavity, the fraction of volume within a distance a of the wall is

$$f_{\text{outer}}(a; R) = 1 - \left(1 - \frac{a}{R}\right)^3, \qquad 0 \le a \le R. \tag{2}$$

Equation (2) was used in the surface-to-volume comparison shown in Supplementary Figure 16.

1.2 Trajectory assembly

LAMMPS custom dumps were parsed frame by frame. The required columns were the molecule identifier, atom type, and Cartesian coordinates `mol type x y z`. Atom type 1 was oxygen and atom type 2 was hydrogen. Each molecule was required to contain exactly one oxygen and two hydrogens. Frames failing this topology check, the expected molecule count, or the expected sampling grid were rejected.

R30 was stored in two trajectory files. A step offset of 550000 was applied to the second segment before sorting and duplicate removal. Thus the two segments were treated as one ordered validation record without altering their coordinates. All molecular and pair observables were recomputed directly from the atomic coordinates; no trajectory-averaged observable was read from a LAMMPS log.

1.3 Two sampling conventions

Two deliberately distinct sampling conventions were retained because they serve different purposes.

Whole-radius validation (Supplementary Figures 1 to 16). The complete endpoints were included. Each NPBC system contributed 201 frames separated by 5 ps over a nominal 1 ns interval: R15 and R30 used 100–1100 ps, whereas R20 used 0–1000 ps. Per-frame temporal standard deviations in these figures were therefore calculated from all 201 frames. Regional block estimates used ten nominal 100 ps blocks; the inclusive final endpoint was assigned to the tenth block.

Matched RDF extension (Supplementary Figures 17 to 21). RDFs were sampled every 25 ps. R15, R20, and R30 each contributed 41 frames, including both endpoints. PBC40 contributed 33 frames over 200–1000 ps. The first temporal standard deviation was calculated from nominal 100 ps RDF blocks.

Final radial-layer analysis (Supplementary Figures 22 to 28). To make every temporal block exactly equal, half-open windows were used:

$$\mathrm{R15} : [100, 1100)\ \mathrm{ps}, \qquad n_f = 200, \tag{3}$$

$$\mathrm{R20} : [0, 1000)\ \mathrm{ps}, \qquad n_f = 200, \tag{4}$$

$$\mathrm{R30} : [100, 1100)\ \mathrm{ps}, \qquad n_f = 200, \tag{5}$$

$$\mathrm{PBC40} : [200, 1000)\ \mathrm{ps}, \qquad n_f = 160. \tag{6}$$

Frames were separated by 5 ps. Each 100 ps block consequently contained 20 frames, giving ten blocks for each NPBC system and eight blocks for PBC40. This half-open convention is the definitive convention for the radial-layer RDFs, shell histograms, and their temporal fluctuation bands.

2 Molecular reconstruction and single-molecule observables

2.1 Molecular center of mass and radial coordinate

With $m_{\mathrm{O}} = 15.999$ and $m_{\mathrm{H}} = 1.008$, the center of mass of water molecule i was

$$\boldsymbol{R}_i^{\mathrm{COM}} = \frac{m_{\mathrm{O}}\boldsymbol{R}_{i\mathrm{O}} + m_{\mathrm{H}}(\boldsymbol{R}_{i\mathrm{H}_1} + \boldsymbol{R}_{i\mathrm{H}_2})}{m_{\mathrm{O}} + 2m_{\mathrm{H}}}. \tag{7}$$

Its absolute radial coordinate and distance from the wall were

$$r_i = \left|\boldsymbol{R}_i^{\mathrm{COM}}\right|, \qquad d_i = R - r_i. \tag{8}$$

For radial-layer assignment only, negative values caused by very small one-step overshoots were clipped to zero. Overshoot counts and maximum overshoots were retained as quality-control observables.

2.2 Dipole, bond, and molecular-plane orientation

The unnormalized molecular dipole-bisector direction was

$$\boldsymbol{b}_i = \frac{\boldsymbol{R}_{i\mathrm{H}_1} + \boldsymbol{R}_{i\mathrm{H}_2}}{2} - \boldsymbol{R}_{i\mathrm{O}}, \tag{9}$$

and the outward radial unit vector was $\hat{\boldsymbol{r}}_i = \boldsymbol{R}_i^{\mathrm{COM}}/r_i$. The primary orientational variable was

$$\cos\theta_i = \hat{\boldsymbol{b}}_i \cdot \hat{\boldsymbol{r}}_i. \tag{10}$$

Legendre order parameters were evaluated molecule by molecule as

$$P_{\ell,i} = P_\ell(\cos\theta_i), \qquad \ell = 1, \ldots, 6. \tag{11}$$

The water-plane normal was

$$\hat{\boldsymbol{n}}_i = \frac{(\boldsymbol{R}_{i\mathrm{H}_1} - \boldsymbol{R}_{i\mathrm{O}}) \times (\boldsymbol{R}_{i\mathrm{H}_2} - \boldsymbol{R}_{i\mathrm{O}})}{|(\boldsymbol{R}_{i\mathrm{H}_1} - \boldsymbol{R}_{i\mathrm{O}}) \times (\boldsymbol{R}_{i\mathrm{H}_2} - \boldsymbol{R}_{i\mathrm{O}})|}, \tag{12}$$

and its second-rank radial order was $P_2(\hat{\boldsymbol{n}}_i \cdot \hat{\boldsymbol{r}}_i)$. Finally, the bond order parameters were the average over the two unit O–H bonds,

$$P_\ell^{\mathrm{bond}}(i) = \frac{1}{2}\sum_{a=1}^{2} P_\ell(\hat{\boldsymbol{e}}_{ia} \cdot \hat{\boldsymbol{r}}_i), \qquad \ell = 1, 2. \tag{13}$$

Undefined directions at the exact cavity center were stored as missing values and excluded only from the affected orientational average. The full-radius dipole profiles are reported in Supplementary Figure 2, and their volume-weighted distributions are included in Supplementary Figure 13.

2.3 Intramolecular geometry

The two O–H lengths were $l_{ia} = |\boldsymbol{R}_{i\mathrm{H}_a} - \boldsymbol{R}_{i\mathrm{O}}|$. The reported mean bond length and H–O–H angle were

$$\bar{l}_{\mathrm{OH},i} = \frac{1}{2}(l_{i1} + l_{i2}), \tag{14}$$

$$\alpha_i = \cos^{-1}(\hat{\boldsymbol{e}}_{i1} \cdot \hat{\boldsymbol{e}}_{i2}). \tag{15}$$

Angles were converted to degrees. The radial means and shell-conditioned distributions are shown in Supplementary Figures 4 and 26, respectively.

2.4 Nearest-neighbor distances and coordination

A Cartesian `cKDTree` of oxygen positions was constructed for each NPBC frame. For every water, the first through fifth distinct oxygen-neighbor distances were retained as $d_{1,\mathrm{OO}}, \ldots, d_{5,\mathrm{OO}}$. The coordination number was the direct count

$$N_{\mathrm{OO},i}(3.5) = \sum_{j \neq i} \Theta(3.5\,\text{Å} - r_{ij}^{\mathrm{OO}}). \tag{16}$$

For PBC40, the same calculation used a periodic `cKDTree` with a 40 Å box and minimum-image distances. The radial first- and fourth-neighbor means are shown in Supplementary Figure 5; the complete first-through-fifth distributions are shown in Supplementary Figure 27.

2.5 Tetrahedral order

Let $\hat{\boldsymbol{u}}_{ij}$ denote the unit vector from oxygen i to one of its four nearest oxygen neighbors. The tetrahedral order parameter was

$$q_i = 1 - \frac{3}{8} \sum_{j=1}^{3} \sum_{k=j+1}^{4} \left(\hat{\boldsymbol{u}}_{ij} \cdot \hat{\boldsymbol{u}}_{ik} + \frac{1}{3} \right)^2 . \tag{17}$$

The periodic calculation applied the minimum-image convention to all four neighbor vectors before evaluating equation (17). Tetrahedrality enters the radial network profiles in Supplementary Figure 3 and the layer-resolved distributions in Supplementary Figure 25.

2.6 Hydrogen bonds

A directional hydrogen bond $i \to j$ was assigned when

$$r_{ij}^{\mathrm{OO}} \leq 3.5\,\text{Å} \quad \text{and} \quad \angle(\mathrm{O}_i - \mathrm{H}_{ia} \cdots \mathrm{O}_j) \geq 150^\circ \tag{18}$$

for either hydrogen a of donor i. Donated and accepted bonds were counted separately, and their sum was reported as the total hydrogen-bond count per molecule. Periodic O–O and O–H vectors were minimum-imaged before the angular test. The resulting hydrogen-bond profiles and distributions are displayed in Supplementary Figures 3 and 25.

3 Radial profiles, density, and temporal fluctuations

3.1 Spherical shells

The complete cavity was partitioned into shells of width $\Delta r = 0.25\,\text{Å}$. Shell s, bounded by r_s^- and r_s^+, had exact volume

$$V_s = \frac{4\pi}{3} \left[(r_s^+)^3 - (r_s^-)^3 \right] . \tag{19}$$

Molecules were assigned by their COM radius. If $n_s(t)$ is the number of molecular centers in shell s at saved frame t, the normalized density was

$$\widetilde{\rho}_s(t) = \frac{n_s(t)}{V_s \rho_0} . \tag{20}$$

The displayed profile is the arithmetic temporal mean of $\widetilde{\rho}_s(t)$. For a generic per-molecule property $A_i(t)$, the per-frame shell mean was

$$A_s(t) = \frac{\sum_i A_i(t)\, \mathbf{1}[i \in s]}{\sum_i \mathbf{1}[i \in s]} . \tag{21}$$

Empty shell frames were stored as missing values rather than as zero. The density and other full-radius shell means obtained from these estimators are plotted in Supplementary Figures 1 to 5.

3.2 First temporal standard deviation

For a shell with n_s^{valid} finite frame values, the first temporal standard deviation was

$$\sigma_{t,s}(A) = \sqrt{\frac{1}{n_s^{\mathrm{valid}} - 1} \sum_t \left[A_s(t) - \overline{A}_s \right]^2} . \tag{22}$$

Every shaded band labelled "$\pm 1\sigma_t$" is $\overline{A}_s \pm \sigma_{t,s}(A)$. It is not a standard error or a confidence interval. The corresponding temporal SEM stored in the tables is $\sigma_{t,s}/\sqrt{n_s^{\mathrm{valid}}}$. The valid-frame

fraction was also recorded; shells occupied in fewer than 50% of frames were explicitly marked in the full-radius plots.

The shell count variance and Fano factor were

$$F_s = \frac{\mathrm{Var}_t[n_s(t)]}{\langle n_s(t)\rangle_t}, \tag{23}$$

using the sample variance with one degree of freedom. The radial fluctuation profiles are collected in Supplementary Figures 10 and 12; all shaded bands in Supplementary Figures 1 to 5 use the same convention.

4 Global translational and angular-isotropy diagnostics

The displacement of the instantaneous water COM cloud from the cavity origin was quantified by

$$\Delta_{\mathrm{COM}}(t) = \left| \frac{1}{N} \sum_{i=1}^{N} \boldsymbol{R}_i^{\mathrm{COM}}(t) \right|. \tag{24}$$

Angular isotropy was measured from

$$\boldsymbol{M}(t) = \frac{1}{N} \sum_i \hat{\boldsymbol{r}}_i \hat{\boldsymbol{r}}_i^{\mathsf{T}}, \qquad \boldsymbol{Q}(t) = \frac{3}{2}\boldsymbol{M}(t) - \frac{1}{2}\boldsymbol{I}. \tag{25}$$

The reported diagnostics were the Frobenius norm $\|\boldsymbol{Q}\|_F = [\mathrm{Tr}(\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{Q})]^{1/2}$ and the three ordered eigenvalues of $\boldsymbol{Q}$. A perfectly isotropic distribution has $\boldsymbol{Q} = \boldsymbol{0}$. Frame means, frame SDs, 100 ps block means, block SEMs, and block-bootstrap pairwise differences were retained. The corresponding global time traces and summary bands are shown in Supplementary Figure 11.

5 Molecule-weighted and shell-volume-weighted distributions

Four radial regions were defined by the shell-midpoint wall distance $d_s = R - r_s^{\mathrm{mid}}$: the complete cavity; interior $d_s \geq 5\,\text{Å}$; outer 5 Å, $d_s < 5\,\text{Å}$; and outer 2 Å, $d_s < 2\,\text{Å}$.

Let h_{sj} be the count in histogram bin j for shell s. The molecule-weighted probability was

$$p_j^{\mathrm{mol}} = \frac{\sum_{s\in\mathcal{R}} h_{sj}}{\sum_{s\in\mathcal{R}} \sum_k h_{sk}}. \tag{26}$$

For volume weighting, each occupied shell histogram was first normalized, $p_{sj} = h_{sj} / \sum_k h_{sk}$, and then averaged with exact shell-volume weights,

$$p_j^{\mathrm{vol}} = \sum_{s\in\mathcal{R}} w_s p_{sj}, \qquad w_s = \frac{V_s}{\sum_{u\in\mathcal{R}}^{\mathrm{occupied}} V_u}. \tag{27}$$

Thus molecule weighting answers "what distribution is observed for a random molecule?", whereas volume weighting gives equal physical-volume importance to the spherical shells and prevents the larger population of outer shells from dominating solely through geometry.

Histogram means and variances were calculated from bin midpoints. The bin definitions are listed in table 2. Regional uncertainty used ten block means. The block SEM was the SD of the ten block means divided by $\sqrt{10}$; the quoted two-sided 95% interval used $t_{0.975,9} = 2.262$.

Table 2: Histogram grids used for molecular observables.

Observable	Range	Bins	Width
$\cos\theta, P_1, P_3, P_5$	$[-1, 1]$	80 uniform	0.025
P_2, P_4, P_6, plane P_2, bond P_2	$[-0.5, 1]$	75 uniform	0.020
bond P_1	$[-1, 1]$	80 uniform	0.025
tetrahedrality q	$[-2, 1]$	120 uniform	0.025
$N_{\mathrm{OO}}(3.5)$	integer centers 0–13	unit-width discrete	1
donated H bonds	integer centers 0–4	unit-width discrete	1
accepted H bonds	integer centers 0–6	unit-width discrete	1
total H bonds	integer centers 0–8	unit-width discrete	1
mean O–H length	$[0.75, 1.25]$ Å	100 uniform	0.005 Å
H–O–H angle	$[80, 130]^\circ$	100 uniform	0.5°
first O–O neighbor	$[2.0, 4.5]$ Å	100 uniform	0.025 Å
second O–O neighbor	$[2.0, 5.0]$ Å	120 uniform	0.025 Å
third O–O neighbor	$[2.0, 5.5]$ Å	140 uniform	0.025 Å
fourth O–O neighbor	$[2.0, 6.0]$ Å	160 uniform	0.025 Å
fifth O–O neighbor	$[2.0, 6.5]$ Å	180 uniform	0.025 Å

The resulting whole-cavity shell-volume-weighted distributions are shown in Supplementary Figure 13.

6 Frozen Gaussian mean-field reconstruction

Each frozen radial bias contained Gaussian amplitudes a_k, layer boundaries, and layer-dependent widths σ_k. Gaussian centers were the boundary midpoints r_k. There were $R/0.25$ Å physical layers (60, 80, and 120 for R15, R20, and R30) and one additional control layer. The potential plotted in the analysis was reconstructed on a 0.01 Å radial grid as

$$U(r) = s \sum_k (a_k - a_0) \exp\left[-\frac{1}{2}\left(\frac{r - r_k}{\sigma_k}\right)^2\right], \qquad s = 0.0005 \text{ eV}. \tag{28}$$

Subtracting a_0 fixes the irrelevant additive coefficient reference. The outward radial force was evaluated analytically, not by finite differences:

$$F_r(r) = -\frac{\mathrm{d}U}{\mathrm{d}r} = s \sum_k (a_k - a_0) \exp\left[-\frac{1}{2}\left(\frac{r - r_k}{\sigma_k}\right)^2\right] \frac{r - r_k}{\sigma_k^2}. \tag{29}$$

Potential and force were stored in meV and meV Å^{-1}, respectively. Both the absolute radius r, normalized radius r/R, and wall distance $R - r$ were retained. Input bias and layer-definition files were identified by SHA-256 checksums. The amplitudes and reconstructed fields appear in Supplementary Figures 6 to 9; Supplementary Figure 10 overlays the wall-aligned potential with the boundary density and its first temporal SD.

7 Boundary-safe site–site RDFs

7.1 General directed-pair estimator

All RDFs used bins of width $\Delta r = 0.05$ Å to a maximum distance of 8.0 Å. The exact volume of bin $j = [r_j^-, r_j^+)$ was

$$\Delta V_j = \frac{4\pi}{3}\left[(r_j^+)^3 - (r_j^-)^3\right]. \tag{30}$$

For reference sites of type α and neighbor sites of type β, the directed-pair estimator was

$$g_{\alpha\beta}(r_j) = \frac{C_{\alpha\beta,j}}{N_\alpha^{\mathrm{ref}}\,\rho_\beta\,\Delta V_j}, \tag{31}$$

where $C_{\alpha\beta,j}$ is the accumulated directed-pair count. Sparse distance matrices from `cKDTree` were used; under PBC, coordinates were wrapped into the 40 Å box and the minimum-image convention was applied internally.

7.2 Geometrically boundary-safe NPBC estimator

For Supplementary Figures 14 and 17 to 20, NPBC reference atoms were restricted to

$$|\boldsymbol{r}_{\mathrm{ref}}| \le R - r_{\max} = R - 8\,\text{Å}. \tag{32}$$

Every complete pair-distance sphere of radius at most 8 Å then lies inside the nominal cavity, so no solid-angle correction is needed. Neighbor atoms were not spatially restricted.

Supplementary Figure 14 used total site–site functions: O–H and H–H therefore retain their intramolecular peaks. The extended functions in Supplementary Figures 17 to 20 are intermolecular only: a pair was retained only if its two atoms had different molecule identifiers. O–O self pairs are therefore excluded automatically. PBC40 used all atoms as reference atoms.

These boundary-safe and extended RDFs used $\rho_\beta = N_\beta/V$. For a same-species distinct-pair function the exact finite-system tail is therefore $1 - 1/N$, rather than exactly unity. The difference is at most approximately 2×10^{-3} for R15 and was retained for continuity with the original estimator. The final radial-layer O–O estimator described below removes this finite-N offset explicitly.

7.3 RDF block fluctuations and comparison metrics

Within each 100 ps block b, an RDF $g_b(r_j)$ was calculated from that block's accumulated numerator and denominator. The displayed first temporal SD and its tabulated SEM were

$$\sigma_{g,j} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}[g_b(r_j) - \overline{g}(r_j)]^2}, \tag{33}$$

$$\mathrm{SEM}_{g,j} = \frac{\sigma_{g,j}}{\sqrt{B}}. \tag{34}$$

NPBC–PBC curve differences were summarized over the structural ranges 2.3–8.0 Å (O–O), 1.3–8.0 Å (O–H), and 1.8–8.0 Å (H–H) using RMSE, MAE, maximum absolute difference, and the 7–8 Å tail mean.

Feature locations were selected on the discrete 0.05 Å grid by searching for the relevant extremum in fixed windows: O–O first maximum 2.4–3.2 Å, first minimum 3.2–4.0 Å, and second maximum 4.0–5.2 Å; O–H hydrogen-bond maximum 1.3–2.2 Å, first minimum 2.2–2.8 Å, and second maximum 2.8–3.8 Å; H–H intermolecular maximum 1.8–2.7 Å and first minimum 2.7–3.3 Å.

8 Radially resolved O–O RDF with spherical edge correction

8.1 Layer definition

For Supplementary Figures 22 to 24, the reference water was assigned by its COM wall distance d_i from equation (8), whereas the pair distance was always the oxygen–oxygen separation. Coarse layers were 0–2, 2–4, 4–6, 6–8, and ≥ 8 Å. Fine layers were 1 Å wide from 0–1 through 11–12 Å, followed by a ≥ 12 Å interior category.

8.2 Analytic accessible-volume denominator

Near a spherical boundary, the full pair shell in equation (30) is not available. Each reference oxygen was therefore normalized by the exact volume of the corresponding pair shell that intersects the nominal spherical cavity.

Let $\mathcal{V}(R, s, d)$ be the intersection volume of a cavity sphere of radius R and a sphere of radius s centered a distance d from the cavity center. It was evaluated as

$$\mathcal{V}(R,s,d) = \begin{cases} 0, & d \geq R+s, \\ \dfrac{4\pi}{3}\min(R,s)^3, & d \leq |R-s|, \\ \dfrac{\pi(R+s-d)^2\left[d^2+2d(s+R)-3(s-R)^2\right]}{12d}, & \text{otherwise.} \end{cases} \tag{35}$$

For oxygen i, whose oxygen coordinate has radius $d_i^{\mathrm{O}} = |\boldsymbol{R}_{i\mathrm{O}}|$, the accessible volume of RDF bin j was

$$A_{ij} = \mathcal{V}(R, r_j^+, d_i^{\mathrm{O}}) - \mathcal{V}(R, r_j^-, d_i^{\mathrm{O}}). \tag{36}$$

The layer-resolved RDF was then

$$g_L^{\mathrm{OO}}(r_j) = \frac{C_{L,j}}{\rho_{\mathrm{distinct}} \sum_{i\in L} A_{ij}}, \qquad \rho_{\mathrm{distinct}} = \frac{N_{\mathrm{O}}-1}{4\pi R^3/3}. \tag{37}$$

Counts are directed: each water in layer L is a reference and every other oxygen within 8 Å is a possible neighbor. Equation (37) both corrects the missing solid angle and makes the finite-N distinct-particle tail exactly consistent with unity for a uniform ideal fluid.

The reflector acts on molecular COM coordinates, whereas equation (35) uses the nominal oxygen-center cavity. Oxygen centers marginally outside R were retained because their intersection volume is well defined. They represented 291/105200 = 0.277% of R15, 462/249600 = 0.185% of R20, and 1033/842200 = 0.123% of R30 reference-water samples. These counts were recorded explicitly as a diagnostic.

For PBC40, the corresponding finite-N estimator used

$$g_{\mathrm{PBC}}^{\mathrm{OO}}(r_j) = \frac{C_j}{N_{\mathrm{O}}[(N_{\mathrm{O}}-1)/V]\Delta V_j}, \tag{38}$$

with minimum-image distances.

8.3 Radial RDF metrics

For every layer, the following quantities were tabulated:

$$\mathrm{RMSE}_{a:b} = \left[\frac{1}{n_{a:b}} \sum_{r_j\in[a,b]} \left(g_L(r_j) - g_{\mathrm{PBC}}(r_j)\right)^2\right]^{1/2}, \tag{39}$$

$$\mathrm{MAE}_{a:b} = \frac{1}{n_{a:b}} \sum_{r_j\in[a,b]} \left|g_L(r_j) - g_{\mathrm{PBC}}(r_j)\right|, \tag{40}$$

using 2.2–6.0 and 2.2–8.0 Å ranges as applicable. The direct coordination number was obtained from counts, rather than by integrating the edge-corrected RDF,

$$\overline{N}_L(3.5) = \frac{\sum_{j:r_j^+\leq 3.5\text{Å}} C_{L,j}}{N_L^{\mathrm{ref}}}. \tag{41}$$

This deliberately preserves the physically missing neighbors at the interface, while equation (37) asks whether correlations in the available volume differ from bulk.

First peak, first minimum, and second peak were searched in 2.4–3.2, 3.2–4.0, and 4.0–5.2 Å, respectively. RDF RMSE and direct coordination were recomputed for every 100 ps block; their first temporal SDs are the uncertainty bands in Supplementary Figure 24.

9 Radial-layer property histograms and PBC comparison

The shell-conditioned histogram analysis used the five coarse wall layers and the following 12 intrinsic quantities: tetrahedrality, O–O coordination, donated, accepted, and total hydrogen bonds, mean O–H length, H–O–H angle, and first through fifth O–O neighbor distances. Radial orientation variables were intentionally excluded because a translationally invariant periodic system has no physical cavity center or radial axis.

For property A, layer L, and bin j, the molecule-weighted probability was

$$p_{L,j}(A) = \frac{h_{L,j}(A)}{\sum_k h_{L,k}(A)}. \tag{42}$$

The same fixed bin edges were used for NPBC and PBC. A histogram was also formed independently in every equal 100 ps block; the per-bin first temporal SD and SEM were calculated across these block probabilities. Supplementary Figures 25 to 27 show the full PBC temporal band and the outer 0–2 Å NPBC temporal band, with the other layer means overlaid.

Distributional differences were quantified by the base-2 Jensen–Shannon distance

$$M = \frac{1}{2}(P + Q), \tag{43}$$

$$D_{\mathrm{JS}}(P, Q) = \left[\frac{1}{2}D_{\mathrm{KL},2}(P\|M) + \frac{1}{2}D_{\mathrm{KL},2}(Q\|M)\right]^{1/2}, \tag{44}$$

and the one-dimensional Wasserstein distance evaluated on histogram bin midpoints. The implementation uses the square-root Jensen–Shannon *distance*, not the unsquared divergence; with base-2 logarithms it lies between zero and one. Means were also evaluated from bin midpoints and reported as $\overline{A}_L - \overline{A}_{\mathrm{PBC}}$. For temporal diagnostics, each NPBC block distribution was compared with the full PBC distribution, and the first SD across block-level distances was stored. The shell-resolved histograms and their aggregate Jensen–Shannon distances are presented in Supplementary Figures 25 to 28.

10 Cross-radius practical-equivalence screen

The practical-equivalence screen was applied separately for each radius pair, region, weighting convention, and observable. It is a conservative multi-criterion screen and not a proof that two probability measures are identical.

Let μ_A, μ_B be block means and s_A, s_B be the SDs of the complete histogram distributions. The pooled distribution SD and standardized mean difference were

$$s_p = \sqrt{\frac{s_A^2 + s_B^2}{2}}, \qquad d = \frac{\mu_A - \mu_B}{s_p}. \tag{45}$$

The Wasserstein distance was normalized as $W_{mathrmn} = W_1(P_A, P_B)/s_p$. A 95% confidence interval for $\mu_A - \mu_B$ was generated from 20000 independent bootstrap resamples of the ten block means of each system, using a fixed reproducible random seed. The raw equivalence margin was $\delta = 0.2 s_p$.

An observable passed the screen only if all four conditions held:

$$|d| \le 0.2, \qquad D_{\mathrm{JS}} \le 0.1, \qquad W_{\mathrm{n}} \le 0.2, \qquad \mathrm{CI}_{95\%}(\mu_A - \mu_B) \subseteq [-\delta, +\delta]. \tag{46}$$

Shell profiles were additionally compared at identical wall-distance bin centers over 0–8, 0–5, 0–2, and 5–10 Å bands using Pearson correlation, RMSE, MAE, and maximum absolute difference. The standardized effect-size matrix and the geometric/equivalence summary are shown in Supplementary Figures 15 and 16.

11 Comparison with experimental water RDF data

11.1 Full experimental O–O curve

The O–O benchmark was the author-supplied numerical curve of Skinner *et al.*[1], measured at 22 °C. The archived table contains 2000 points on a 0.01 Å grid from 0 to 19.99 Å, together with a pointwise estimated error. Simulation curves were compared on their 0.05 Å bin centers over the common 1.1–7.975 Å range. The experimental curve and error were linearly interpolated onto those centers.

For simulation value $g_s(r_j)$, experimental value $g_e(r_j)$, and experimental estimated error $\epsilon_e(r_j)$, the residual and descriptive combined standard uncertainty were

$$\Delta_j = g_s(r_j) - g_e(r_j), \qquad u_{c,j} = \sqrt{\epsilon_e(r_j)^2 + \mathrm{SEM}_{s,j}^2}. \tag{47}$$

RMSE, MAE, mean signed difference, maximum absolute difference, RMS residual normalized by ϵ_e, the fraction within ϵ_e, and the fraction within u_c were reported both over the full common range and over the structural range beginning at 2.2 Å. These are descriptive diagnostics, not a formal goodness-of-fit test, because the experimental and simulation state points are not matched in number density and differ slightly in temperature. The complete overlay and simulation-minus-experiment residuals are shown in Supplementary Figure 21.

11.2 Published RDF landmarks

The O–O first maximum, first minimum, and second maximum were additionally compared with the landmarks reported by Skinner *et al.*[1]. Intermolecular O–H and H–H feature landmarks were taken from the joint X-ray/neutron EPSR benchmark of Soper and Benmore[2], as tabulated by Zheng *et al.*[3]. No digitized continuous O–H or H–H experimental curve was introduced; those comparisons are landmark-only. The landmark comparisons are shown in Supplementary Figures 17 to 19, while the direct matched-state NPBC–PBC comparison is given in Supplementary Figure 20.

12 Graphical and numerical output conventions

All figures in Supplementary Figures 1 to 28 follow the graphics convention pinned to STOCHKIN commit `3c99adb`; the complete 40-character commit identifier is stored in the archived suite metadata. Each numbered figure was exported as a 300 dpi PNG and a vector PDF. PNG headers and minimum sizes were checked, and every PDF was validated with `pdfinfo`. Unless stated otherwise, a shaded simulation band denotes the first temporal SD, not SEM.

The main figure-to-estimator mapping is:

Table 3: Computation represented by each supplementary figure.

Figure	Quantity and estimator
1	Full-radius COM density from equation (20); per-frame mean and first temporal SD.
2	Dipole P_1 and P_2 from equation (11).
3	Tetrahedrality, direct coordination, and total H-bond profiles from equations (16) to (18).
4	Mean O–H length and H–O–H angle from equation (15).
5	First and fourth O–O nearest-neighbor profiles.
6	Frozen physical Gaussian amplitudes over the absolute radius; control layer shown separately.
7	Same amplitudes aligned by normalized radius r/R.

Figure	Quantity and estimator
8	Exact potential and analytic conservative force from equations (28) and (29).
9	Same potential and force aligned by wall distance $R - r$.
10	Boundary density mean $\pm 1\sigma_t$ overlaid with $U(R - r)$.
11	Global COM shift and $\|\boldsymbol{Q}\|_F$ from equations (24) and (25).
12	First temporal SD of density and network observables as a function of r/R.
13	Whole-cavity shell-volume-weighted histograms from equation (27).
14	Boundary-safe total O–O, O–H, and H–H RDFs from equations (31) and (32).
15	Cross-radius standardized mean differences from equation (45).
16	Exact outer-volume fractions and fraction of tests passing equation (46).
17–19	Intermolecular RDF curves, block SDs, and experimental feature landmarks.
20	Matched-state NPBC–PBC intermolecular RDF residuals; residual band is the quadrature sum of NPBC and PBC block SDs.
21	Full O–O experimental overlay and residuals from equation (47).
22	Coarse wall-layer edge-corrected O–O RDFs from equation (37) and residuals versus PBC.
23	Fine 1 Å heat map of $g_L^{\mathrm{OO}} - g_{\mathrm{PBC}}^{\mathrm{OO}}$.
24	Fine-layer RDF RMSE, direct coordination, peak height, and minimum height.
25	Radial network-property histograms from equation (42) versus PBC.
26	Radial intramolecular-geometry histograms versus PBC.
27	First–fifth O–O-neighbor histograms versus PBC.
28	Base-2 Jensen–Shannon distance from equation (44) for all 12 intrinsic distributions.

13 Quality assurance and reproducibility

The following checks were applied before packaging the results:

1. Every dump frame was parsed with strict water stoichiometry and expected molecule counts (526, 1248, 4211, and 2383 waters).

2. Duplicate global steps were removed after applying the documented R30 offset, and each analysis required its exact expected frame count.

3. Trajectories, frozen biases, layer-definition files, the PBC model, and the experimental RDF table were identified by SHA-256 checksums.

4. The radial-layer analysis recovered 105200, 249600, 842200, and 381280 reference-water samples for R15, R20, R30, and PBC40, respectively.

5. Coarse and fine radial RDF partitions contained identical total pair counts for each NPBC system.

6. All 192 system–layer–property histograms normalized to unity within floating-point tolerance, and all 54 layer-level RDF metric rows were finite.

7. The analytic accessible volumes were checked to lie between zero and the corresponding full spherical-shell volume.

8. All 28 PNG/PDF figure pairs in Supplementary Figures 1 to 28 were regenerated from the packaged tables; the figure manifest maps every image to its caption.

9. A package-level SHA-256 manifest covers every archived analysis file.

The numerical outputs required to reproduce the figures include the complete frame-by-shell table, shell temporal statistics, regional distributions and equivalence tests, exact potential amplitudes and fields, boundary-safe and periodic RDF counts, radial-layer RDF numerators and denominators, block SDs and SEMs, radial histograms, distribution-distance metrics, source-file inventories, and analysis metadata. The scripts `make_stochkin_figure_suite.py`, `compute_rdf_extensions.py`, and `compute_radial_layer_analysis.py` regenerate the analysis from the archived inputs.

14 Interpretive limitation

The edge-corrected RDF and direct coordination answer complementary questions. The RDF divides by the locally accessible pair-shell volume and therefore measures correlations in directions available to the liquid; the direct coordination retains the physically absent neighbors beyond the wall. A large direct coordination loss together with a more modest edge-corrected RDF change is therefore not contradictory. Likewise, whole-cavity distributions mix interfacial and interior water in a radius-dependent proportion given by equation (2); they should not be interpreted as pure bulk distributions. The radial-layer analysis is the appropriate estimator for separating a localized interfacial response from an altered cavity interior.

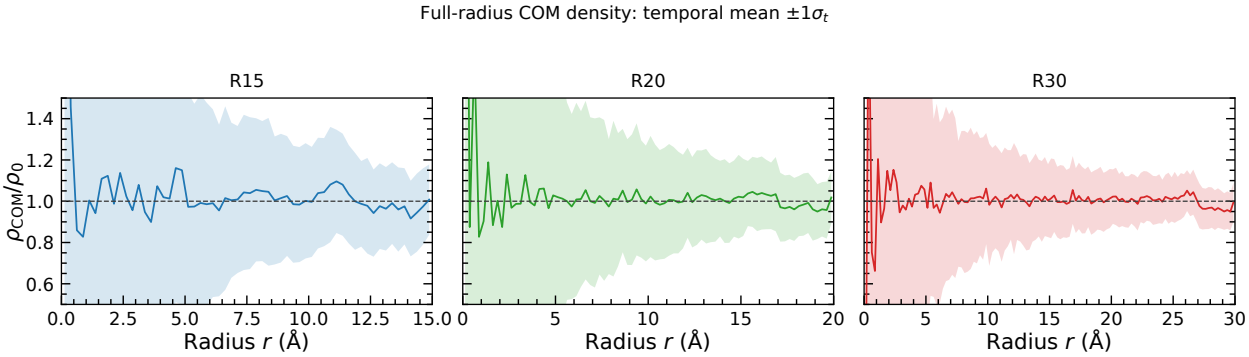
References

[1] L. B. Skinner, C. Huang, D. Schlesinger, L. G. M. Pettersson, A. Nilsson, and C. J. Benmore, "Benchmark oxygen–oxygen pair-distribution function of ambient water from x-ray diffraction measurements with a wide (Q)-range," *J. Chem. Phys.* **138**, 074506 (2013). doi:10.1063/1.4790861.

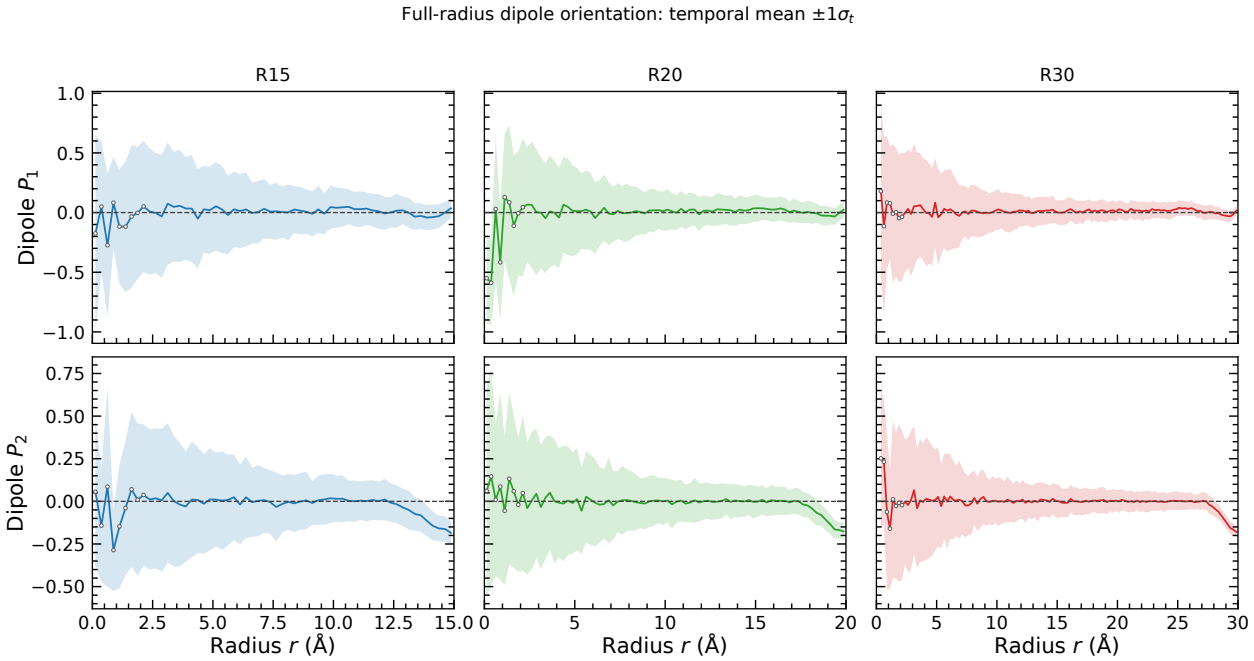
[2] A. K. Soper and C. J. Benmore, "Quantum differences between heavy and light water," *Phys. Rev. Lett.* **101**, 065502 (2008). doi:10.1103/PhysRevLett.101.065502.

[3] L. Zheng, M. Chen, Z. Sun, H.-Y. Ko, B. Santra, P. Dhuvad, and X. Wu, "Structural, electronic, and dynamical properties of liquid water by ab initio molecular dynamics based on SCAN functional within the canonical ensemble," *J. Chem. Phys.* **148**, 164505 (2018). doi:10.1063/1.5023611.

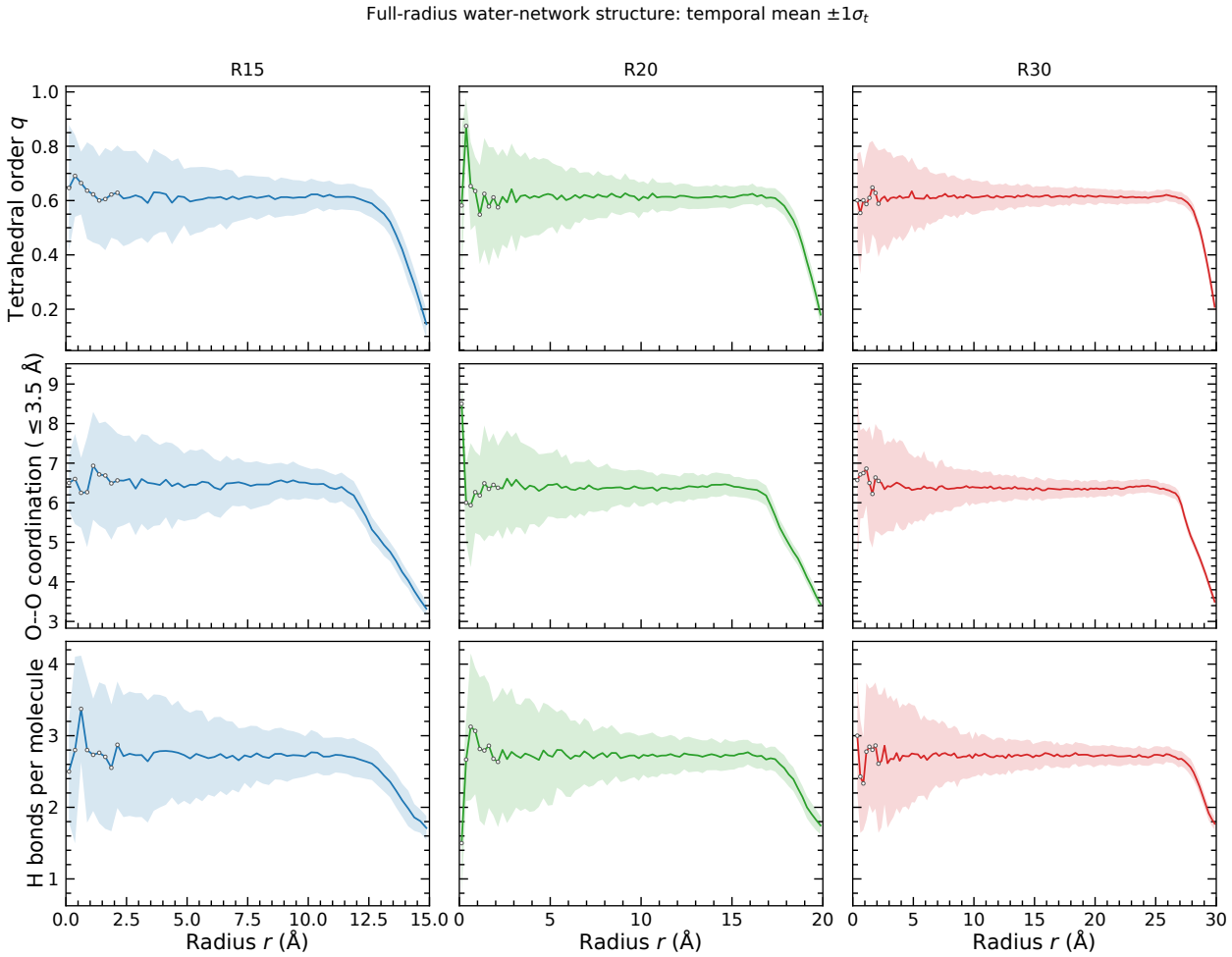
A Supplementary Figures



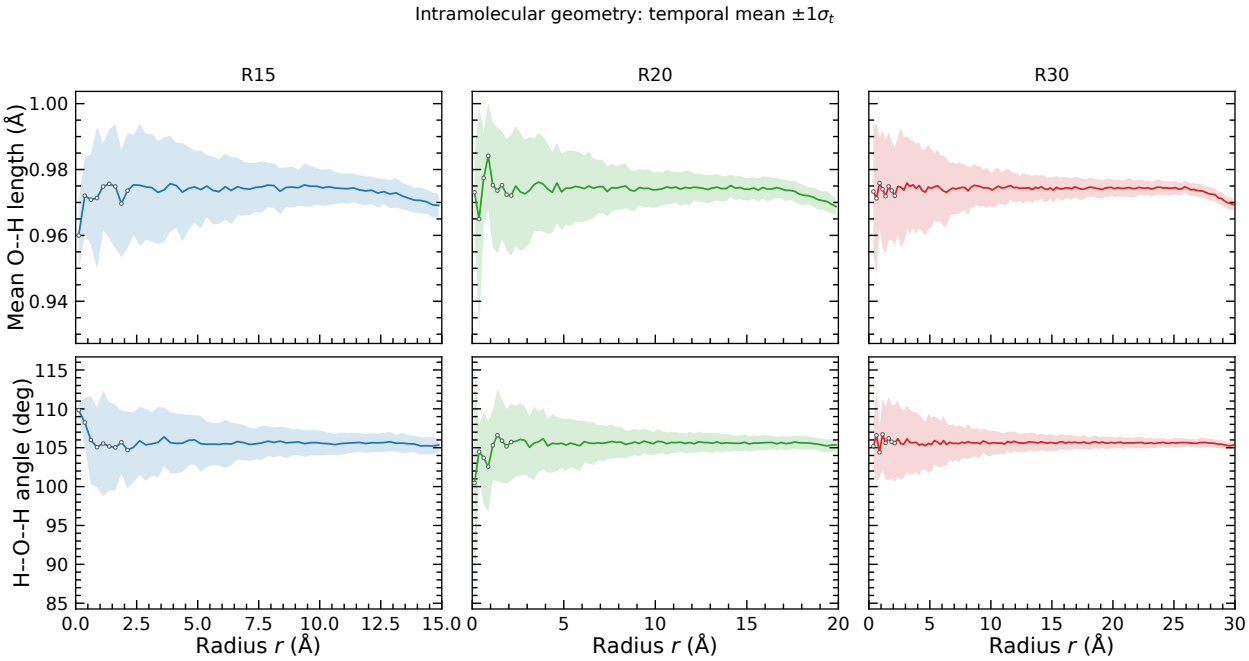
Supplementary Figure 1: Full-radius molecular-COM density. Solid lines are temporal means and shaded bands are the first temporal standard deviation across saved frames.



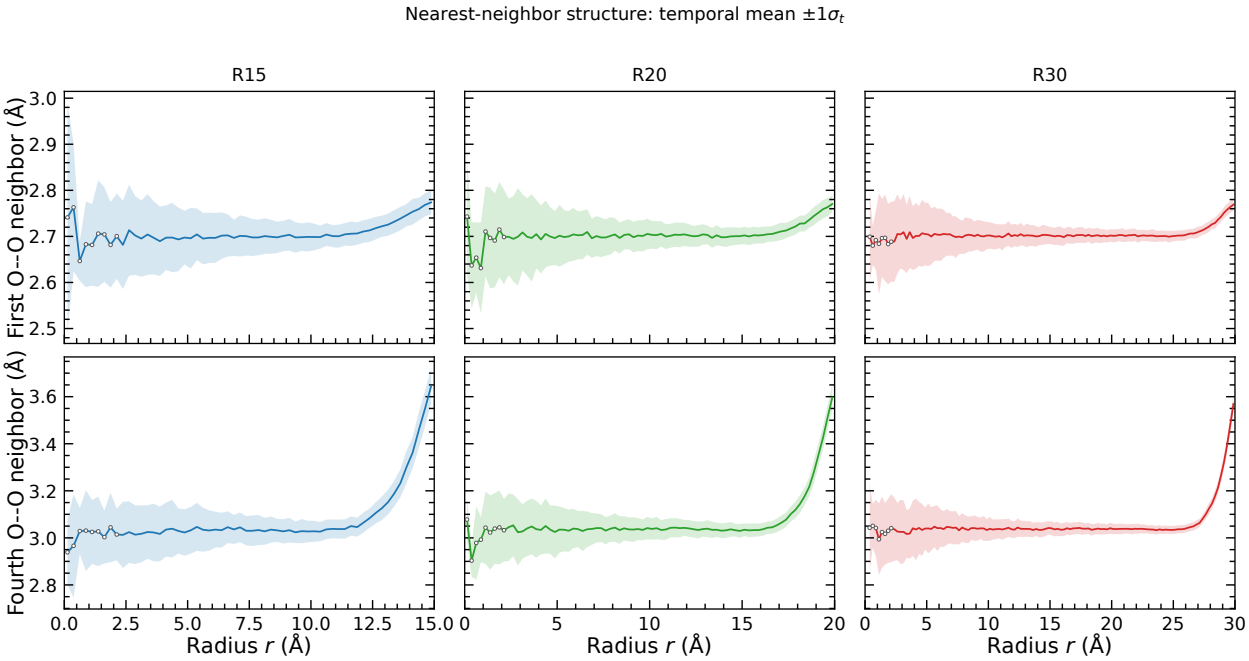
Supplementary Figure 2: Full-radius dipole P_1 and P_2 orientational profiles. Shaded bands denote the first temporal standard deviation.



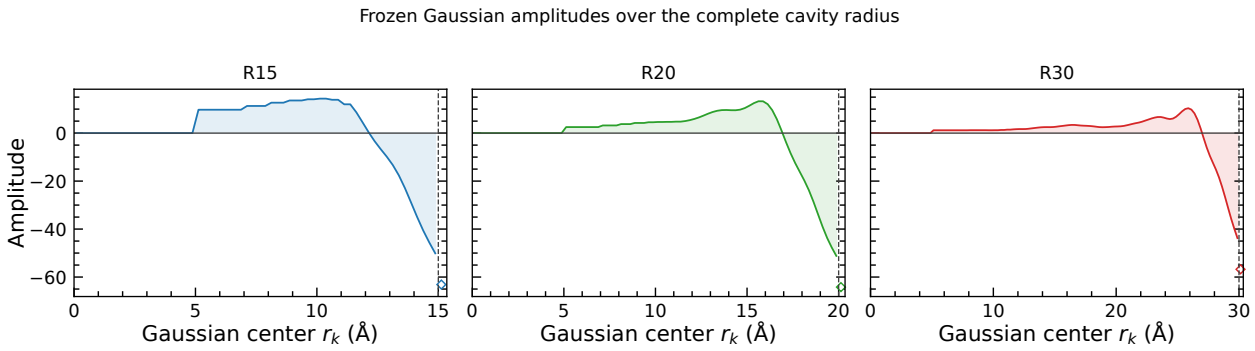
Supplementary Figure 3: Full-radius tetrahedrality, direct O–O coordination, and hydrogen bonding profiles.



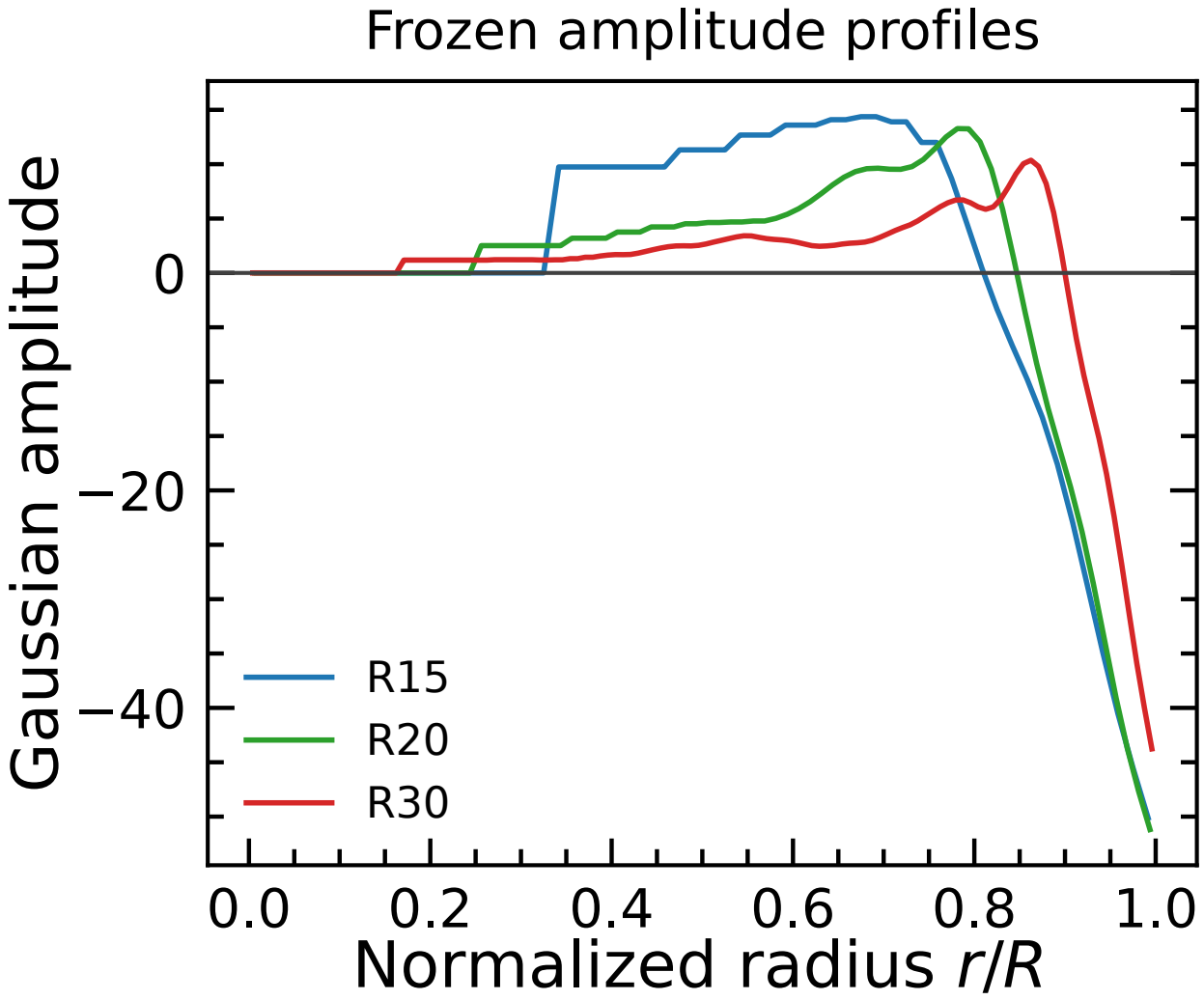
Supplementary Figure 4: Full-radius mean O–H bond length and H–O–H angle profiles.



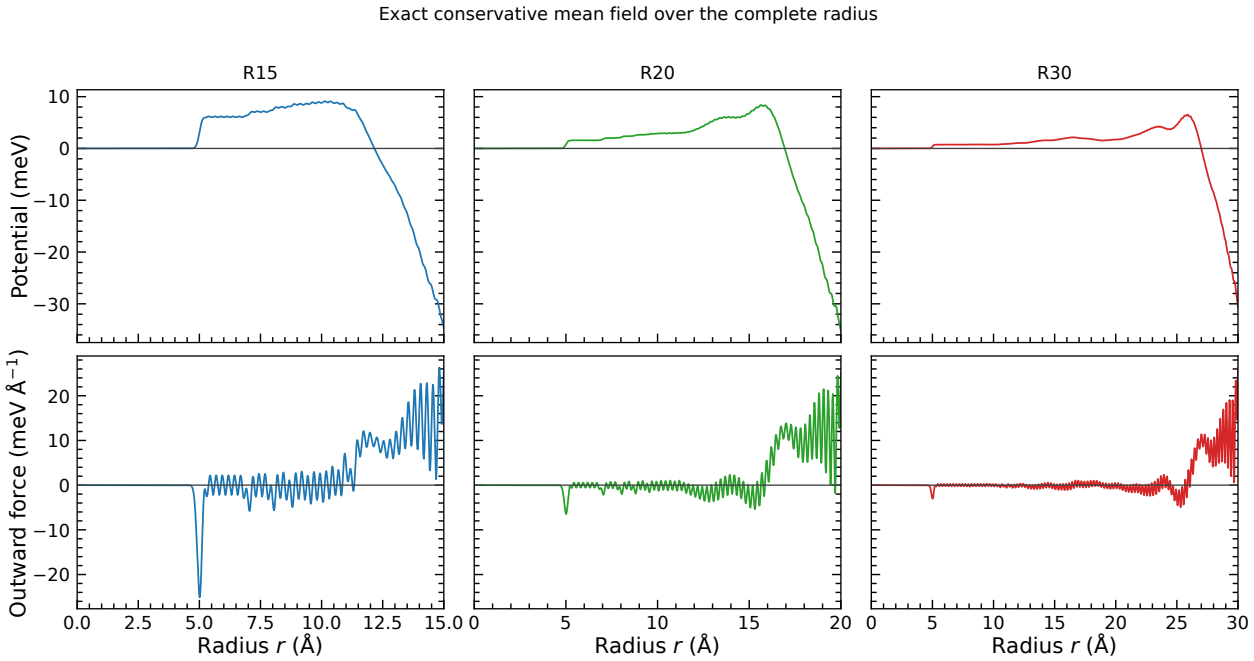
Supplementary Figure 5: Full-radius first- and fourth-nearest O–O neighbor-distance profiles.



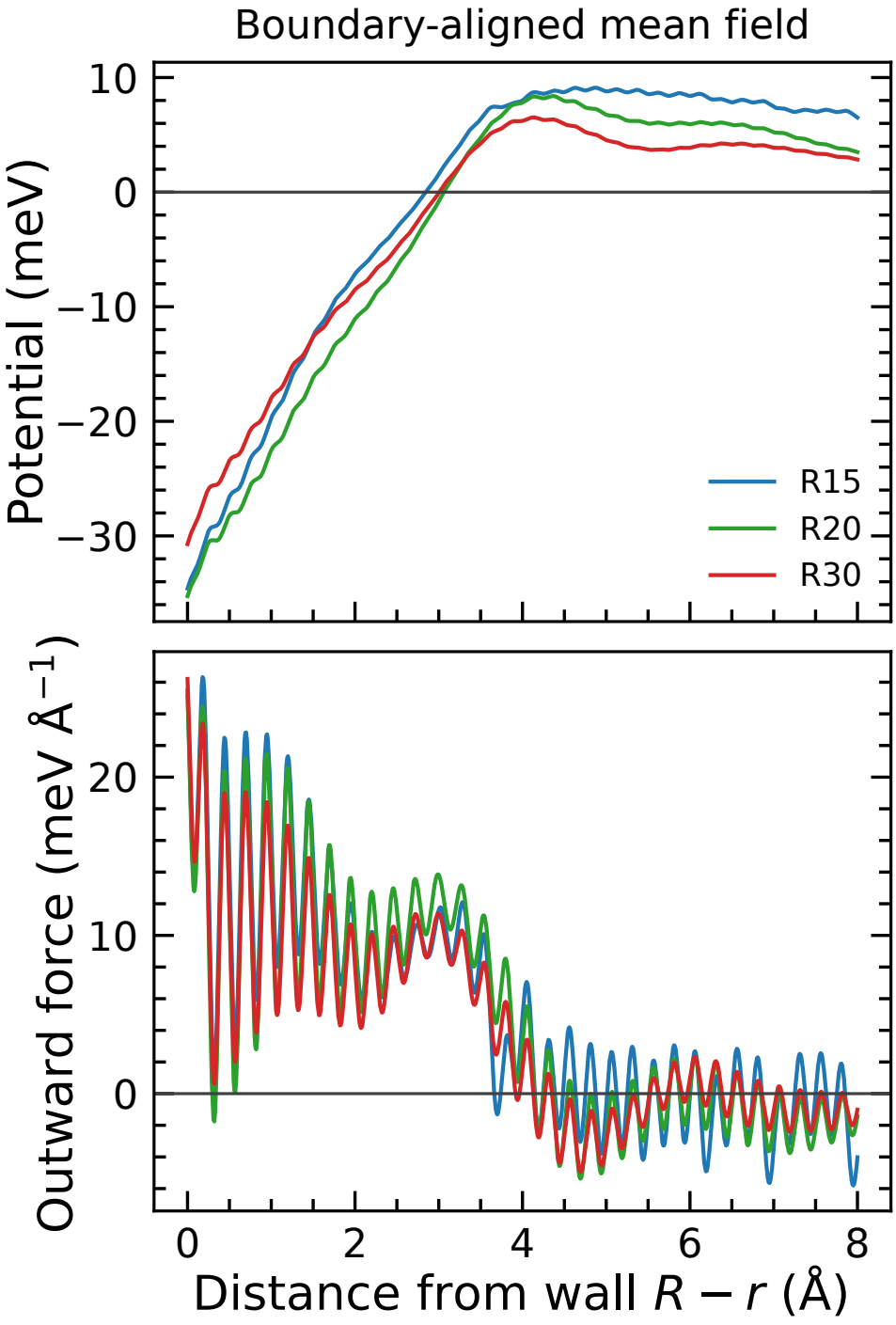
Supplementary Figure 6: Frozen physical Gaussian coefficients over the complete absolute radius. The open diamond denotes the additional control layer.



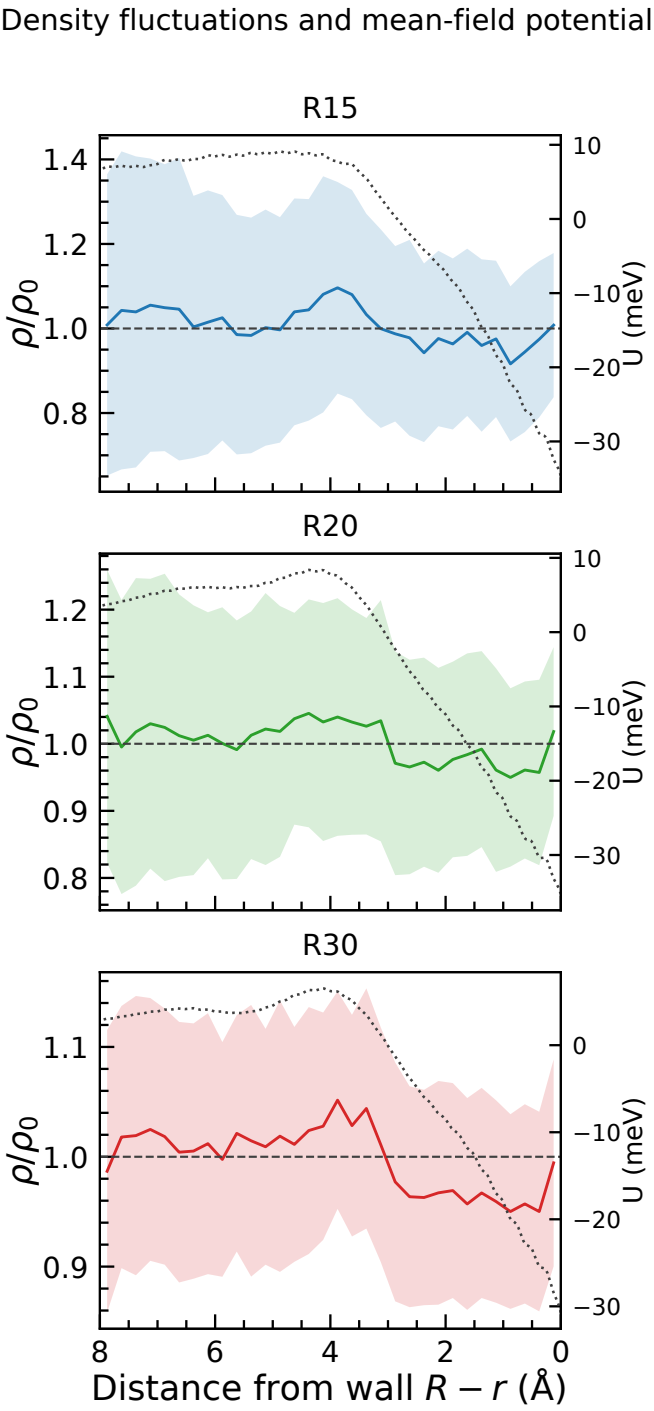
Supplementary Figure 7: Frozen physical Gaussian coefficients aligned by normalized radius r/R.



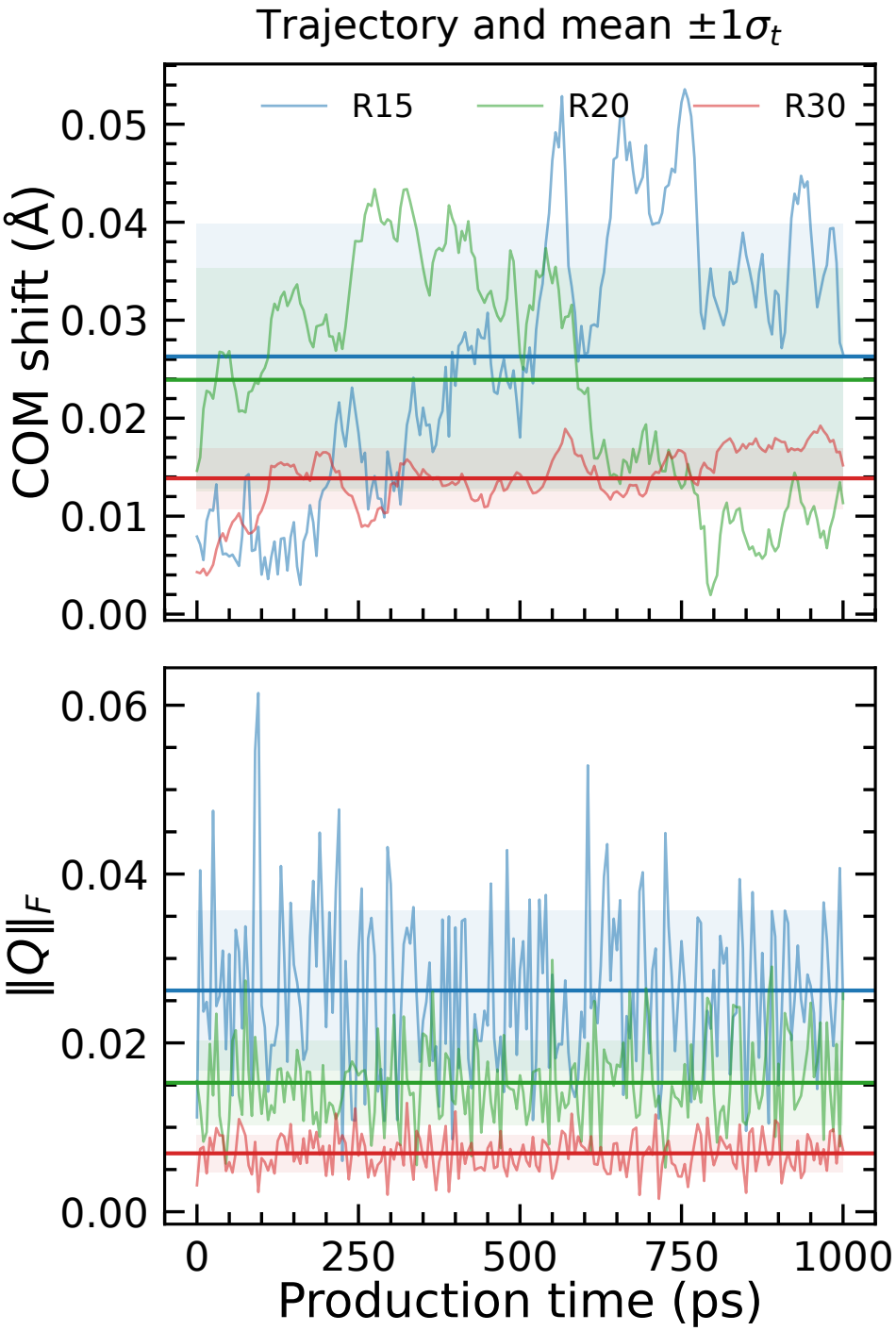
Supplementary Figure 8: Exact Gaussian-sum potential and its analytic conservative radial force over the complete interval $0 \leq r \leq R$.



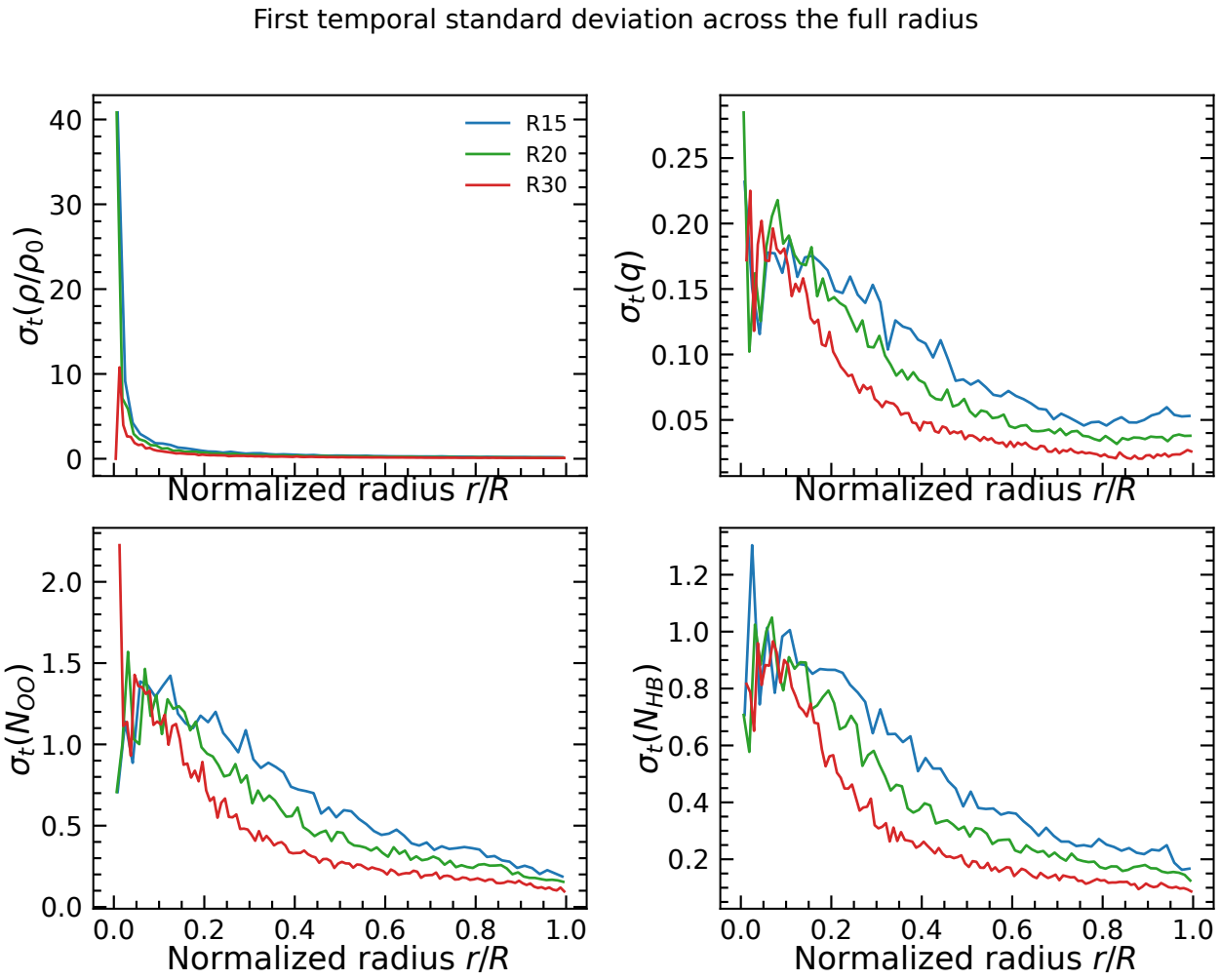
Supplementary Figure 9: Mean-field potential and force aligned by distance from the cavity wall.



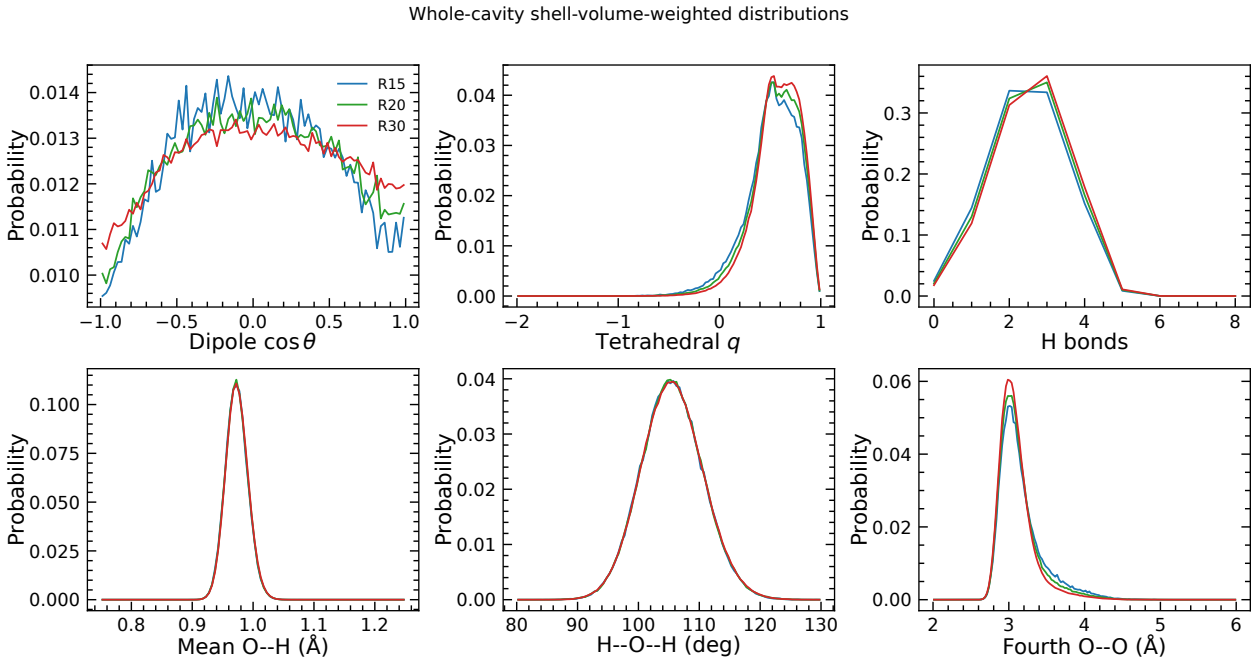
Supplementary Figure 10: Boundary density mean and first temporal standard deviation overlaid with the exact mean-field potential (dotted curve, right axis).



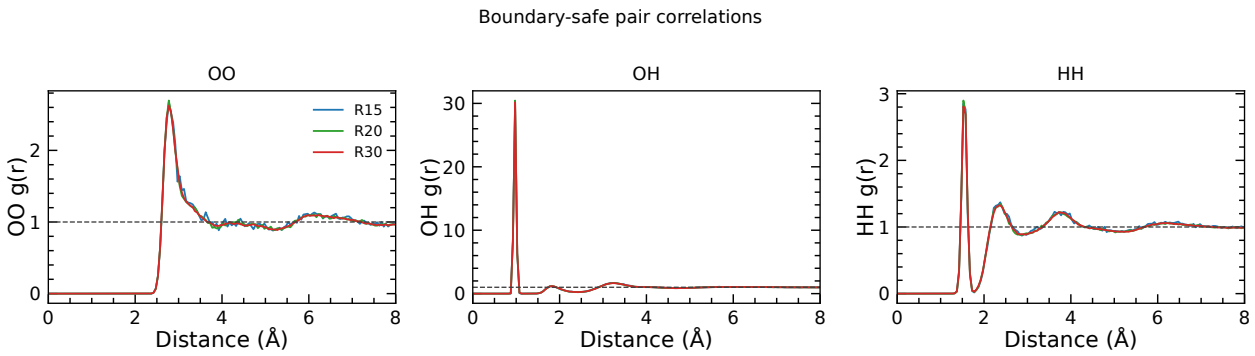
Supplementary Figure 11: Global molecular-COM displacement and angular-anisotropy time traces, with temporal mean and first standard deviation.



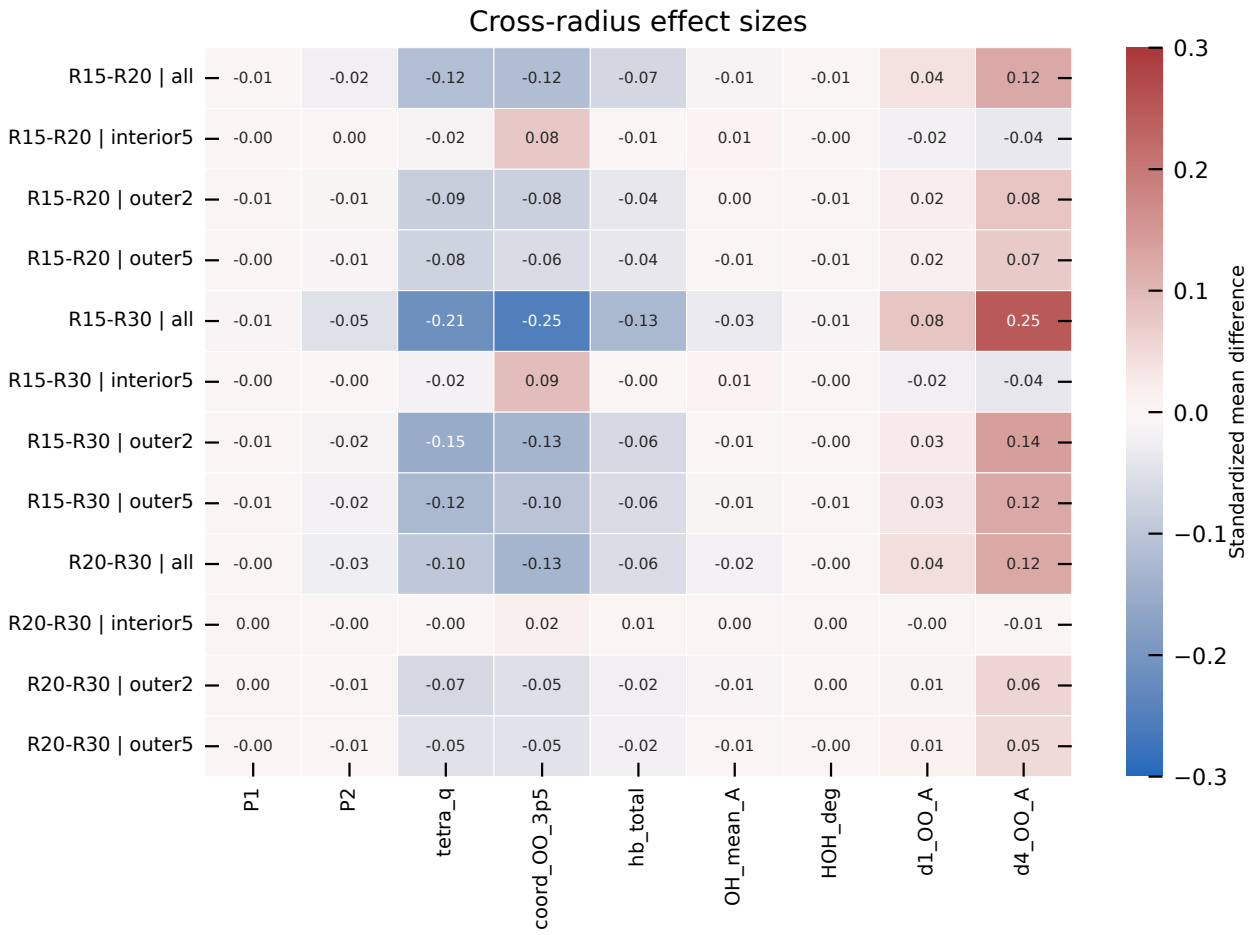
Supplementary Figure 12: First temporal standard deviation of density and network observables over the complete normalized radius.



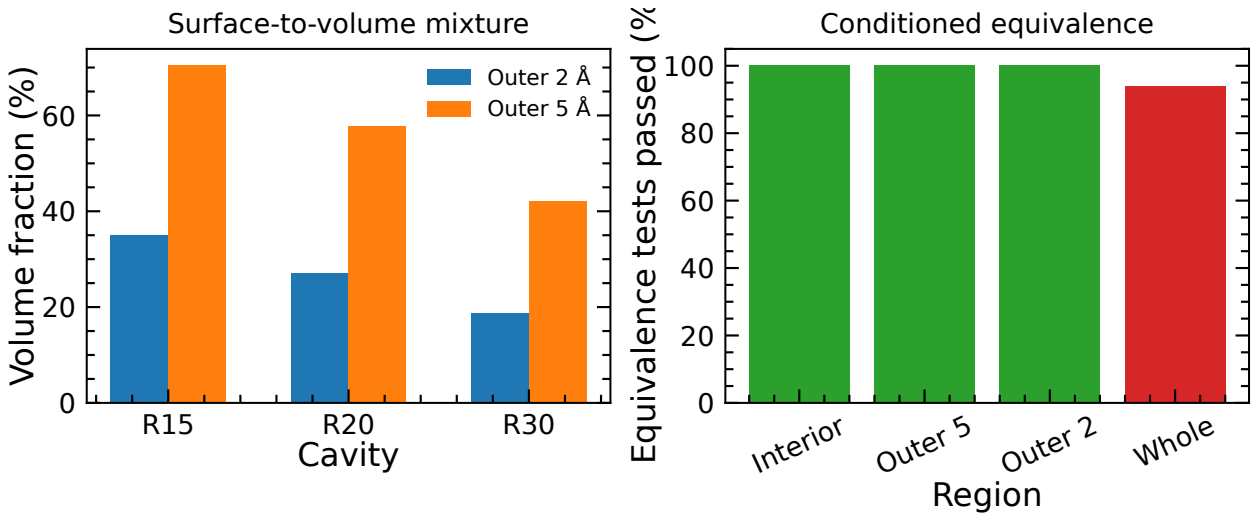
Supplementary Figure 13: Whole-cavity shell-volume-weighted distributions of molecular coordinates and structural observables.



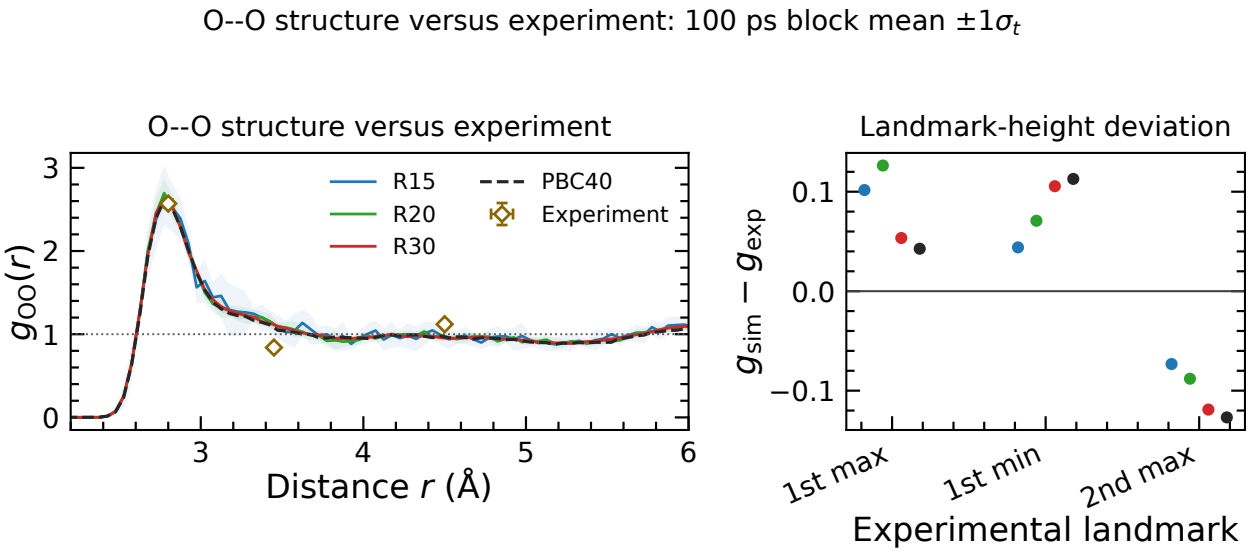
Supplementary Figure 14: Boundary-safe total O–O, O–H, and H–H radial distribution functions. Intramolecular contributions are retained for O–H and H–H.



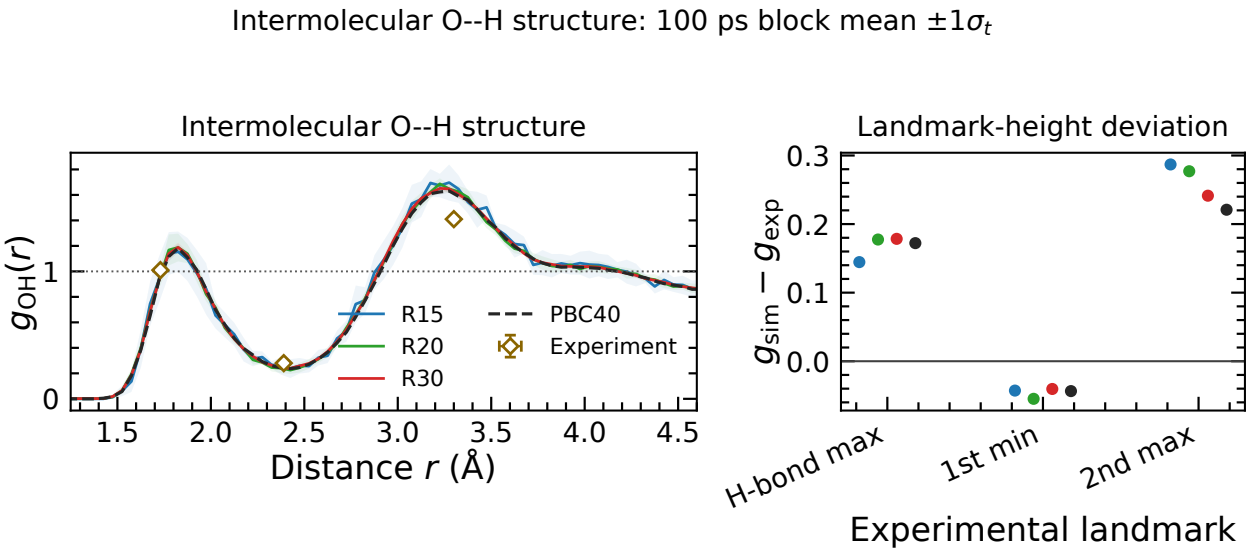
Supplementary Figure 15: Shell-volume-weighted standardized mean differences between cavity radii, resolved by region and observable.



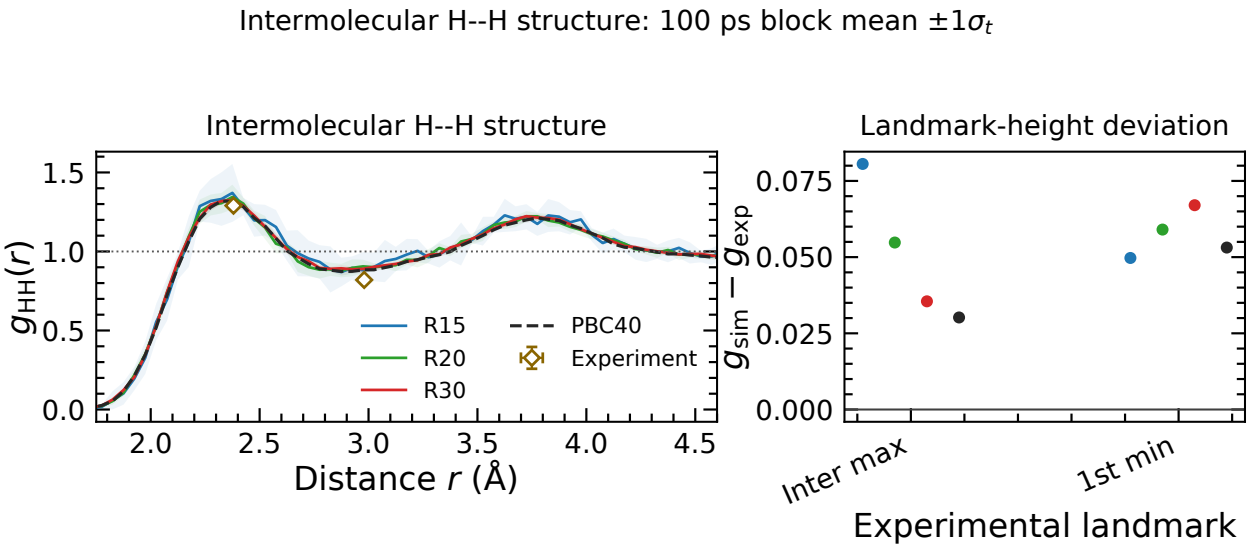
Supplementary Figure 16: Exact surface-volume fractions and the fraction of practical- equivalence tests satisfying all four criteria.



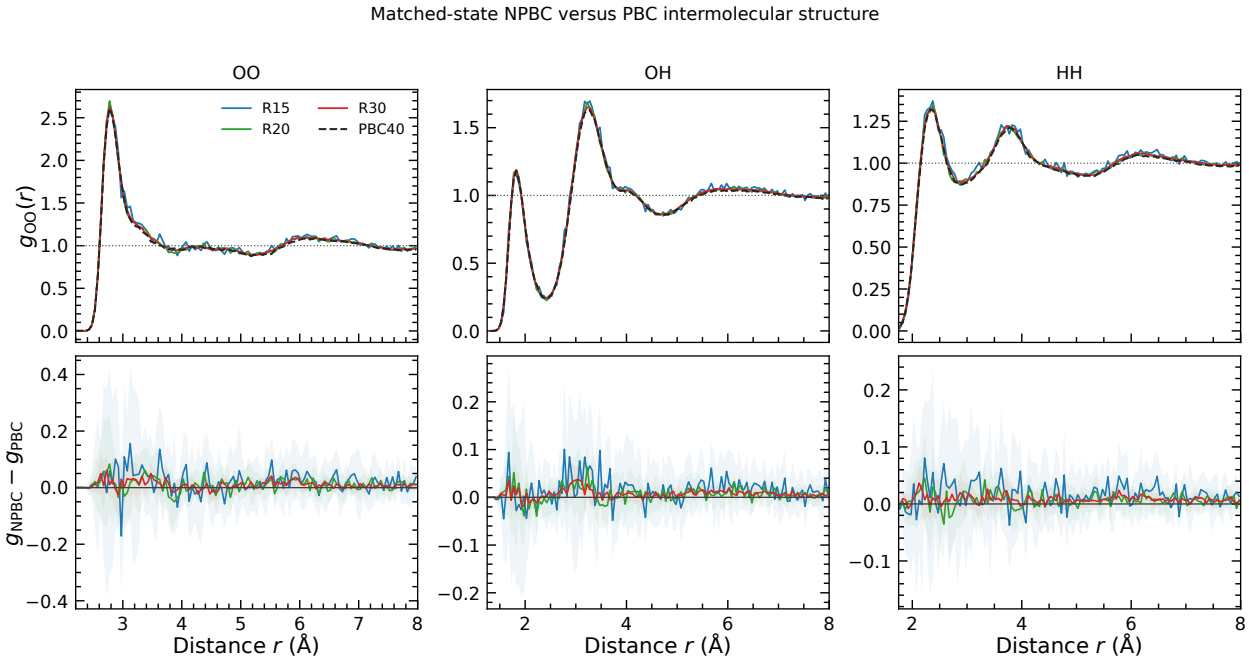
Supplementary Figure 17: Intermolecular O–O structure compared with experimental feature landmarks. Same-molecule pairs are excluded where applicable; shading is the first temporal standard deviation across 100 ps RDF blocks.



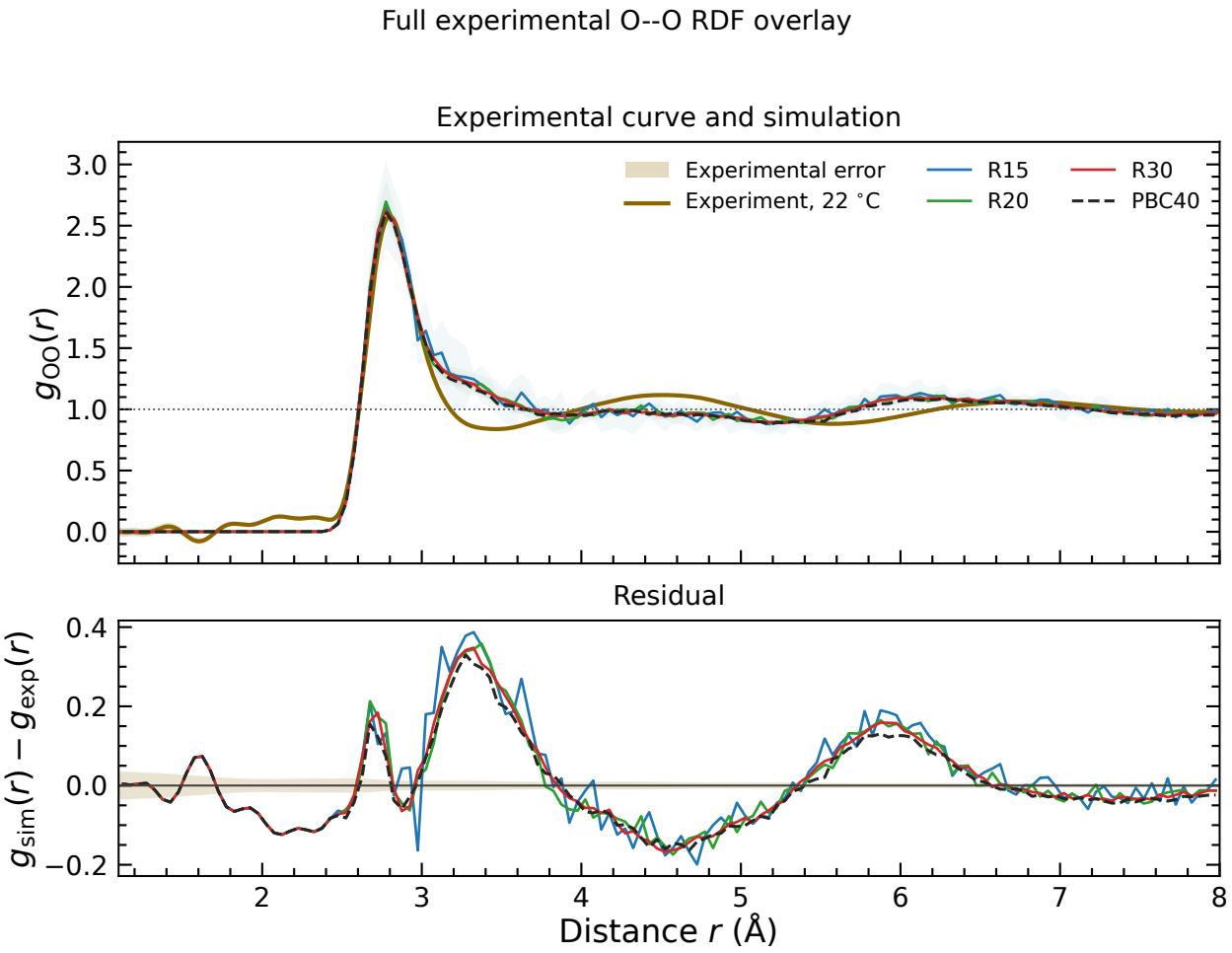
Supplementary Figure 18: Intermolecular O–H structure compared with published feature landmarks. Same-molecule pairs are excluded; shading is the first temporal standard deviation across 100 ps RDF blocks.



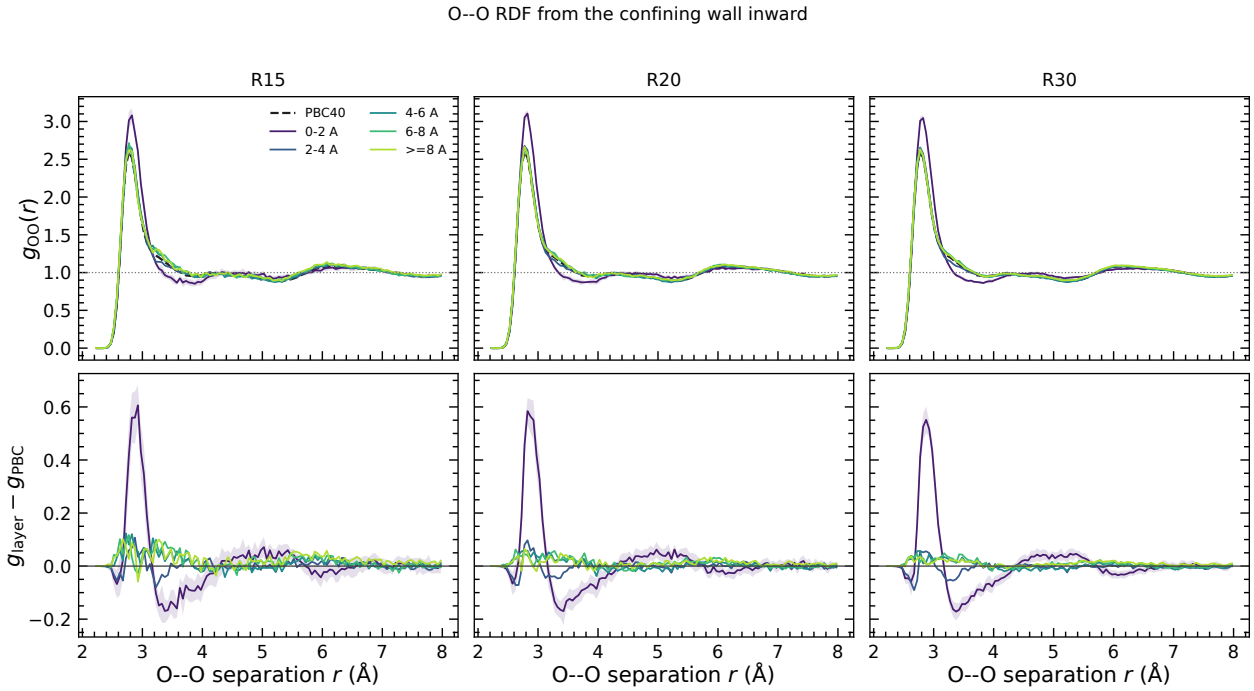
Supplementary Figure 19: Intermolecular H–H structure compared with published feature landmarks. Same-molecule pairs are excluded; shading is the first temporal standard deviation across 100 ps RDF blocks.



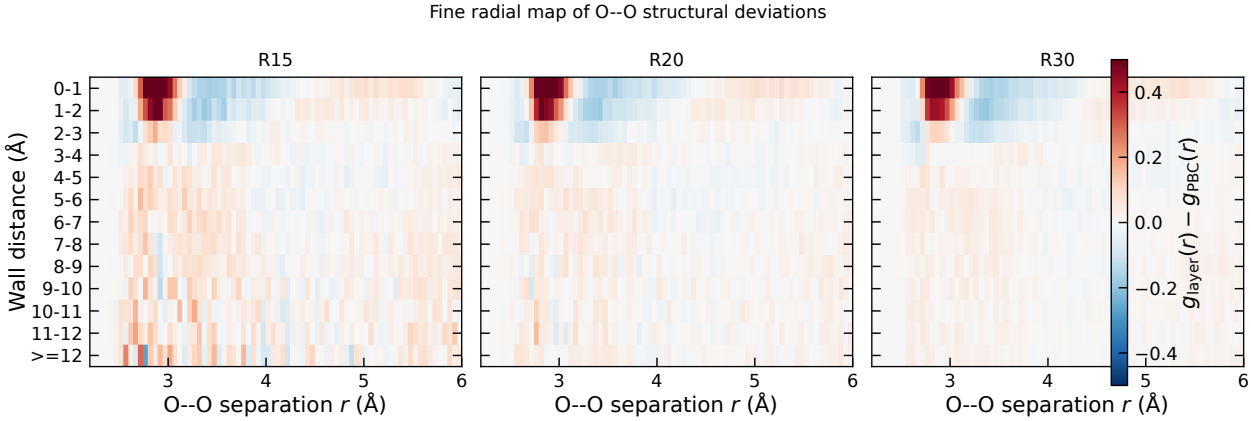
Supplementary Figure 20: Direct comparison of R15, R20, and R30 with the matched 40 Å periodic MACE–OFF23 reference. Residual shading is the quadrature-combined first temporal standard deviation of the NPBC and PBC 100 ps block RDFs.



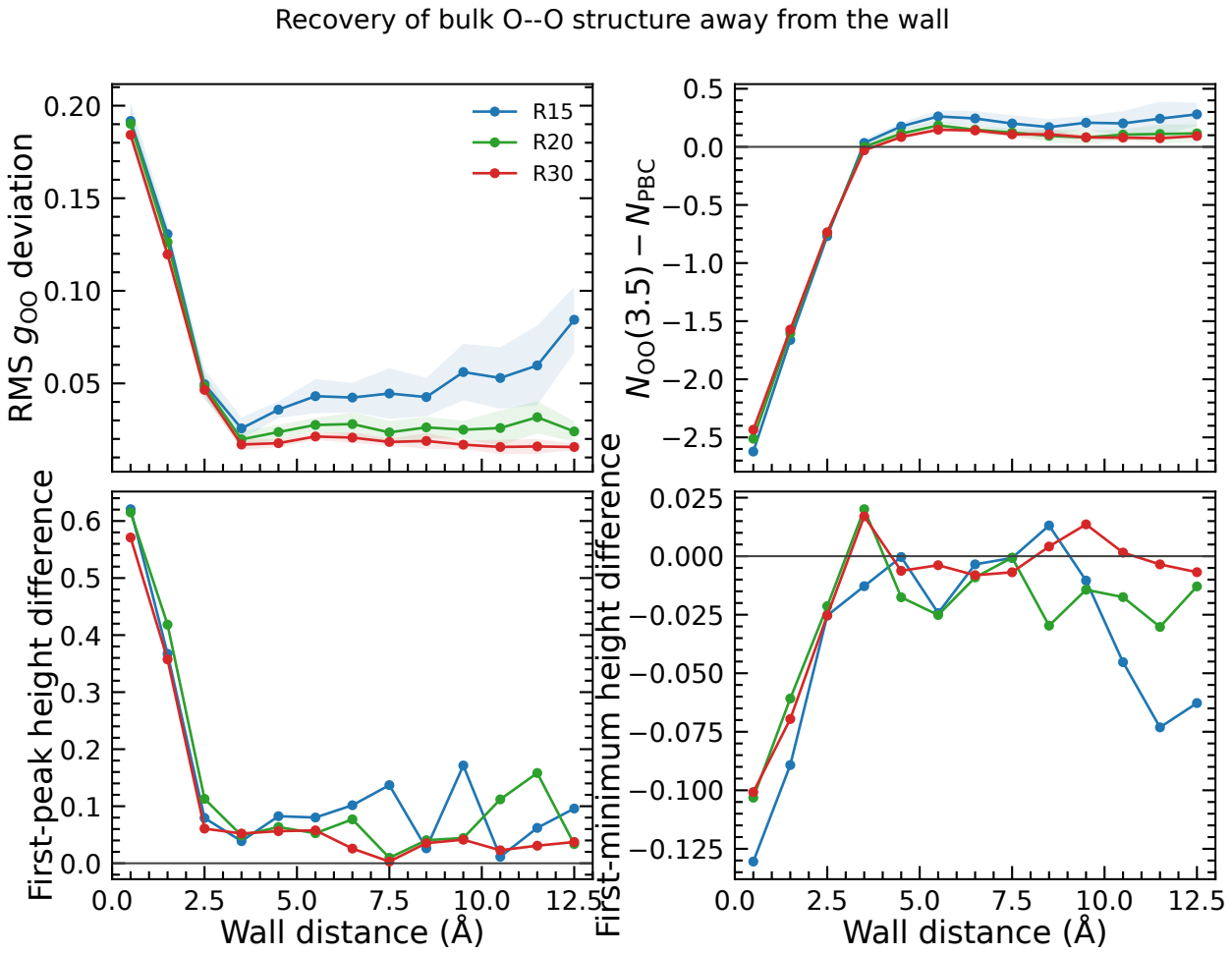
Supplementary Figure 21: Full common-range O–O RDF overlay for R15, R20, R30, and PBC40 with the Skinner *et al.* experimental curve and its pointwise estimated error. Simulation bands are first temporal standard deviations across 100 ps blocks; the lower panel is simulation minus experiment.



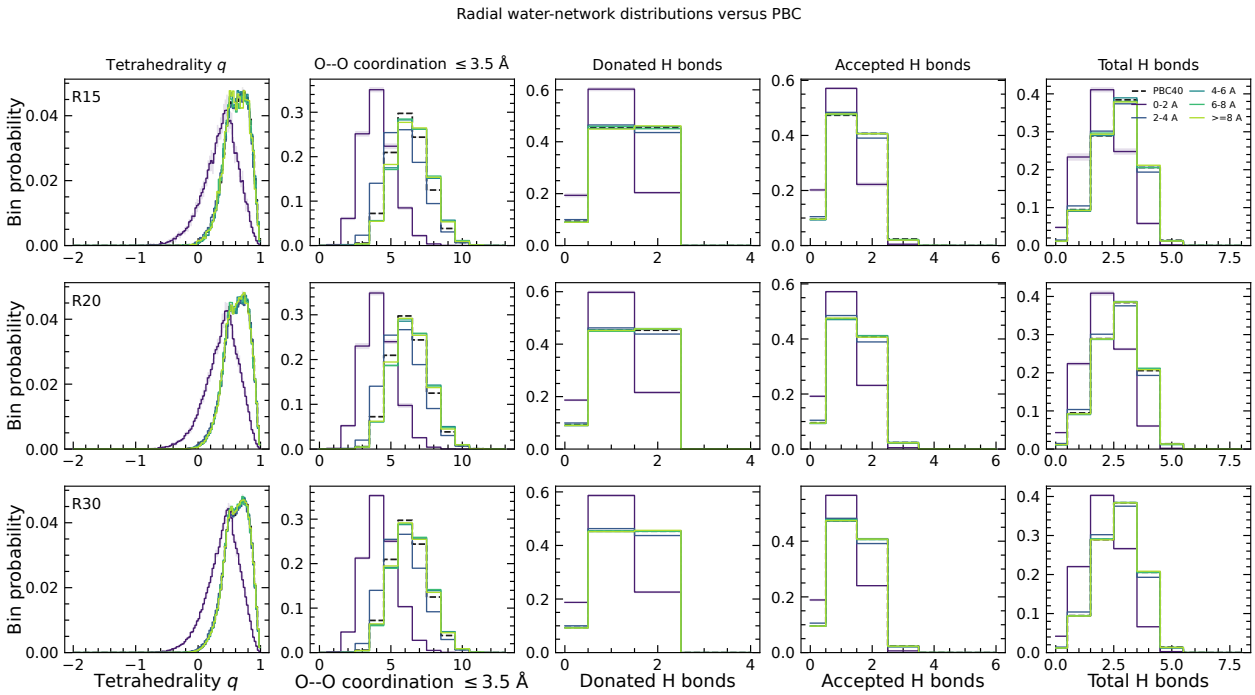
Supplementary Figure 22: O–O RDF conditioned on reference-water COM wall distance. NPBC curves use the exact spherical accessible-shell-volume correction and distinct-particle finite-N normalization. Shading is the first temporal standard deviation across equal 100 ps blocks for PBC and the outer 0–2 Å layer.



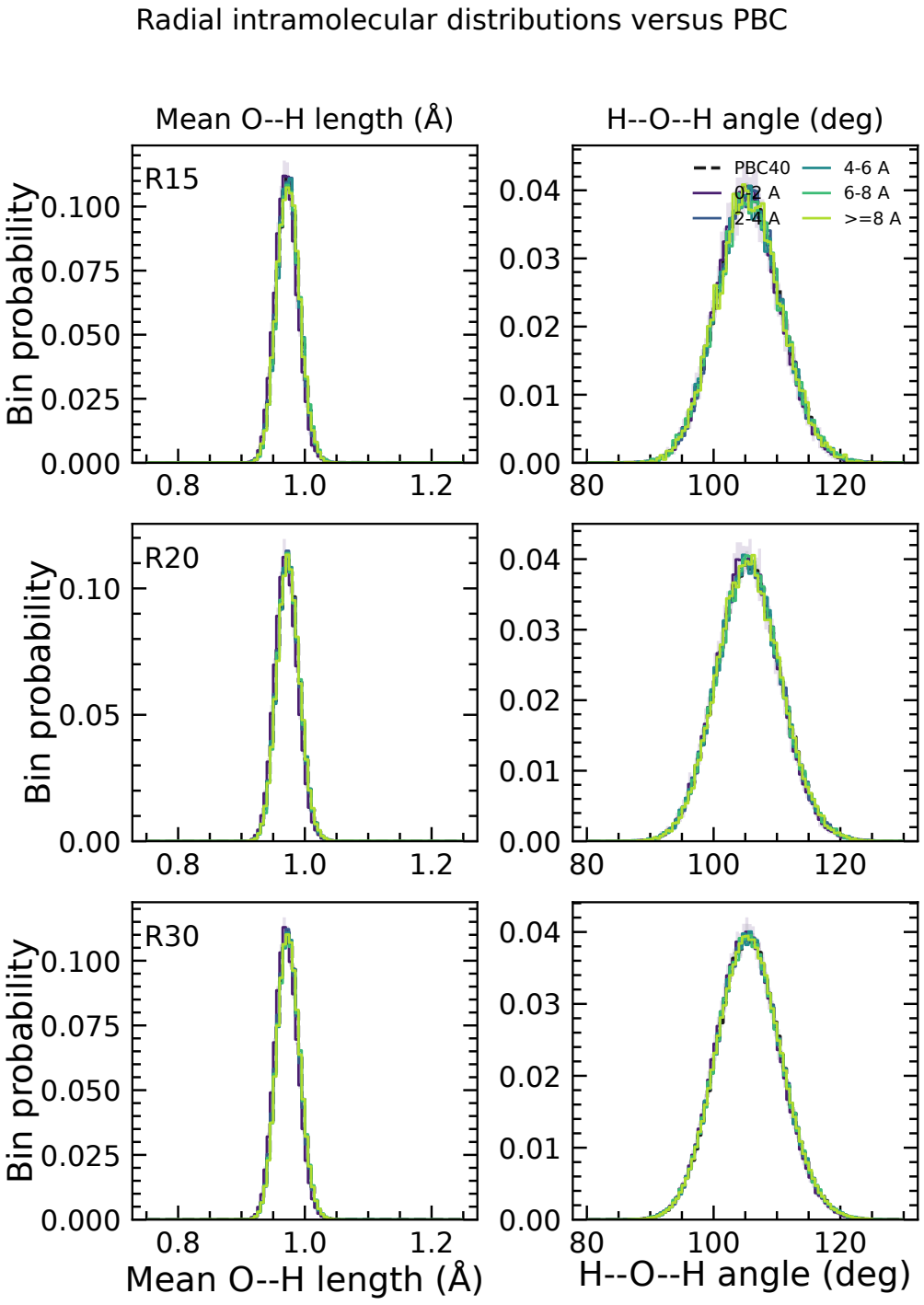
Supplementary Figure 23: Fine 1 Å wall-layer heat map of the edge-corrected $g_{\mathrm{OO},L}(r) - g_{\mathrm{OO,PBC}}(r)$. Rows run from the outer wall inward.



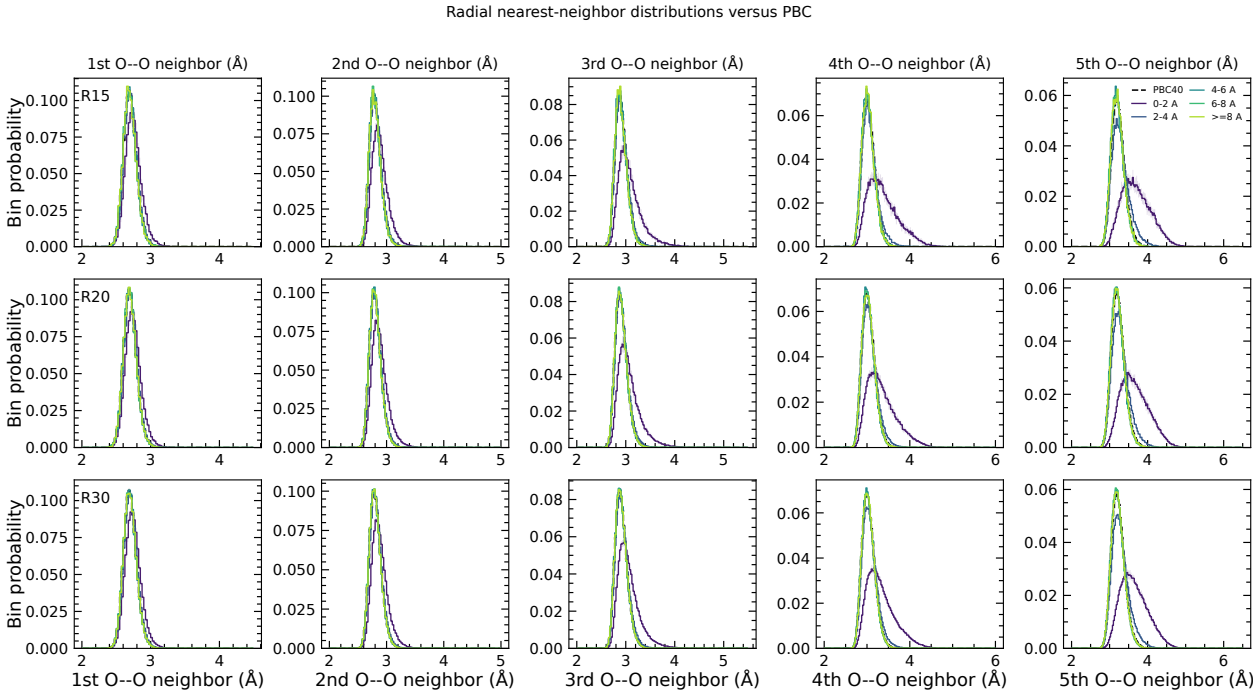
Supplementary Figure 24: Fine-layer RDF deviation metrics versus distance from the wall. Shading on RDF RMSE and direct coordination is the first temporal standard deviation across equal 100 ps blocks.



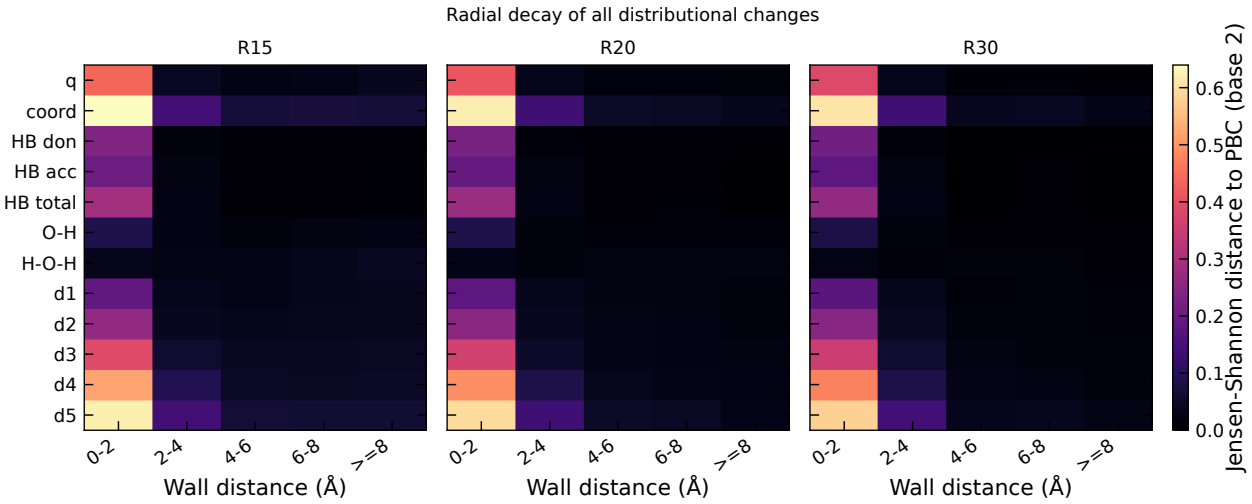
Supplementary Figure 25: Radial water-network distributions compared with PBC. Histograms are molecule-weighted in 2 Å wall-distance layers. Shading is the first temporal standard deviation across equal 100 ps blocks for PBC and the outer 0–2 Å layer.



Supplementary Figure 26: Radial intramolecular O–H length and H–O–H angle distributions compared with PBC. Histograms are molecule-weighted in 2 Å wall-distance layers; shading follows the convention in Supplementary Figure 25.



Supplementary Figure 27: First-through-fifth O–O-neighbor distributions by radial layer and PBC. Histograms are molecule-weighted in 2 Å wall-distance layers; shading follows the convention in Supplementary Figure 25.



Supplementary Figure 28: Base-2 Jensen–Shannon distance between each shell-conditioned intrinsic-property histogram and the matched PBC40 histogram. Zero denotes identical binned distributions.